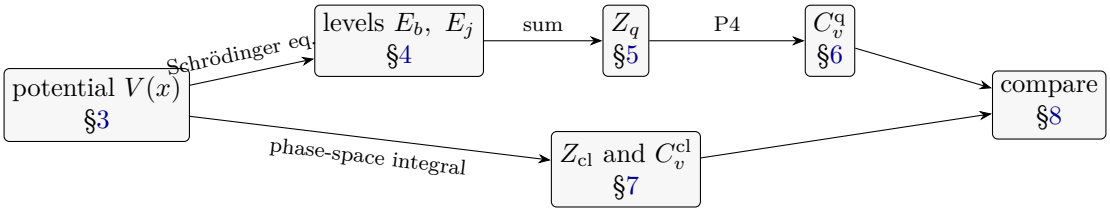
Pöschl–Teller Well in a Box

Every step of the derivation of C_v^{q} and C_v^{cl}, from Calculus 1–2 and linear algebra

Nadav Jean · 28.10.2026 · companion to `PT_Box_Summary.pdf` and `PT_Box_Derivation.pdf`

How to read this

Prerequisites. Calculus 1–2 and linear algebra. Every other tool is built in §1 (mathematics, facts F1–F15) and §2 (physics, P1–P5); later steps cite these labels.
Two routes to the energy levels. The main text solves the Schrödinger equation by factorization ("SUSY", §4.1–4.5). §4.7 derives the same solutions with associated Legendre functions, and §4.8 compares the two and recommends one.
Labels under a step. **Source.** The statement (or its result) is in that source, on the page given. Every page was checked against the copy listed in the References. **Own step.** Constructed for this project by Claude, the AI assistant that wrote the earlier derivation. "No reference found" means no published source was found for it. "(standard)" means Claude's write-up of a textbook argument; the result itself is cited where a page-checked source exists.
Checks. (Cn) is block n of `check_explained.py` in this folder. It recomputes every number quoted here in about 1 min; its output is in `results/check_explained.log`.
Units. Temperatures are given as k_BT in units of $\hbar^2\alpha^2/2m$, and box sizes as αL.
Not covered. Homogeneous strokes and the engine (Derivation §7).



Assumptions

A1 One particle of mass m in one dimension, in equilibrium with a heat bath at temperature T (canonical ensemble), $\beta = 1/k_BT$.

A2 The Pöschl–Teller well with $\lambda = 2$, Eq. (3.2), inside hard walls at $x = \pm L/2$.

A3 $\alpha L \gg 1$. Terms of relative size $e^{-\alpha L}$ are dropped; for $\alpha L \geq 100$ these are below 4×10^{-44}.

A4 $L \gtrsim 2\lambda_{\mathrm{th}}$. The sum-to-integral remainder R of §5 is dropped.

Apart from A3–A4, the closed forms (5.6), (6.4) and (7.8) are exact. Both are checked against exact level sums (C11, C13, C14).

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1 Mathematical toolkit

F1 Hyperbolic functions. $\cosh u = \dfrac{e^u + e^{-u}}{2}$, $\sinh u = \dfrac{e^u - e^{-u}}{2}$, $\tanh u = \dfrac{\sinh u}{\cosh u}$.

(a) $\cosh^2 u - \sinh^2 u = 1$, hence $1 - \tanh^2 u = \dfrac{1}{\cosh^2 u}$. Proof: $(e^u + e^{-u})^2 - (e^u - e^{-u})^2 = 4$. Divide by 4, then by $\cosh^2 u$.

(b) $(\cosh u)' = \sinh u$, $(\sinh u)' = \cosh u$, $(\tanh u)' = \dfrac{\cosh^2 u - \sinh^2 u}{\cosh^2 u} = \dfrac{1}{\cosh^2 u}$ (quotient rule, then (a)).

(c) cosh is even; sinh and tanh are odd (replace u by $-u$ in the definitions).

(d) $\sinh 2u = 2\sinh u\cosh u$ and $\cosh^2 u = \frac{1}{2}(1+\cosh 2u)$ (expand both sides).

(e) For $u > 0$: $0 < 1-\tanh u = \dfrac{2}{e^{2u}+1} < 2e^{-2u}$ and $0 < \dfrac{1}{\cosh^2 u} < 4e^{-2u}$. So as $u \to \pm\infty$, $\tanh u \to \pm 1$ and $1/\cosh^2 u \to 0$, exponentially fast. Proof: $1-\tanh u = \frac{\cosh u - \sinh u}{\cosh u} = \frac{e^{-u}}{\cosh u}$, and $\cosh u > \frac{1}{2}e^{u}$.

(f) $\cosh u - \tanh u \sinh u = \dfrac{1}{\cosh u}$ (by (a)), and $\tanh u \cosh u = \sinh u$.

Source. [DLMF] 4.28.1, 4.28.2, 4.28.4 (definitions); 4.35.11–12 (a); 4.34.1–3 (b); 4.35.23–25 (c); 4.35.26–27 (d).

Own step. (standard) (e) and (f) follow from the definitions; no source checked.

F2 arctan and tan. $\arctan : \mathbb{R} \to (-\frac{\pi}{2}, \frac{\pi}{2})$ is the inverse of tan on $(-\frac{\pi}{2}, \frac{\pi}{2})$.

(a) $\arctan(-u) = -\arctan u$, and $(\arctan u)' = \dfrac{1}{1+u^2}$. Hence, for $c > 0$ (substitute $u = cv$): $\displaystyle\int_0^\infty \frac{\mathrm{d}u}{u^2+c^2} = \frac{1}{c}\big[\arctan v\big]_0^\infty = \frac{\pi}{2c}$.

(b) For $u > 0$: $\arctan\frac{1}{u} = \frac{\pi}{2} - \arctan u$. Proof: by (a), the sum $\arctan\frac{1}{u} + \arctan u$ has derivative $-\frac{1/u^2}{1+1/u^2} + \frac{1}{1+u^2} = 0$, so it is constant. At $u = 1$ it equals $\frac{\pi}{4} + \frac{\pi}{4}$.

(c) tan has period π and is one-to-one on each period: $\tan\varphi_1 = \tan\varphi_2 \iff \varphi_1 = \varphi_2 + n\pi$, $n \in \mathbb{Z}$.

(d) $\tan(\varphi + \frac{\pi}{2}) = -\dfrac{1}{\tan\varphi}$, because $\sin(\varphi+\frac{\pi}{2}) = \cos\varphi$ and $\cos(\varphi+\frac{\pi}{2}) = -\sin\varphi$.

Source. [DLMF] 4.23.12 and 4.24.9 (a); 4.23.9 with 4.23.18 (b); 4.14.10 (c).

F3 Functions as vectors, operators as linear maps. The functions on an interval form a vector space. The maps $f \mapsto f'$, $f \mapsto W(x)f$ and $f \mapsto -\frac{\hbar^2}{2m}f'' + V(x)f$ are linear; physicists call them *operators*. A product of operators is their composition, $(ST)f = S(Tf)$. Composition is associative, but in general $ST \neq TS$, exactly as for matrices. The equation $H\psi = E\psi$ with $\psi \neq 0$ is an eigenvalue equation, the analogue of $Au = \lambda u$.

Lemma (intertwining). If $ST = TR$ and $R\varphi = E\varphi$, then $S(T\varphi) = E\,(T\varphi)$. Proof: $S(T\varphi) = (ST)\varphi = (TR)\varphi = T(R\varphi) = T(E\varphi) = ET\varphi$. □ So, when $T\varphi \neq 0$, it is an eigenvector of S with the same eigenvalue. This is the same mechanism by which similar matrices share eigenvalues.

Source. $H\psi = E\psi$ as the analogue of $Au = \lambda u$: [Tong-QM, p. 68].

Own step. (standard) The lemma.

F4 All solutions of the three free equations. On an interval containing 0, with $k > 0$ and $\kappa > 0$:

(a) $\varphi'' = -k^2\varphi \iff \varphi = C_1\cos kx + C_2\sin kx$; (b) $\varphi'' = 0 \iff \varphi = C_1 + C_2 x$;

(c) $\varphi'' = \kappa^2\varphi \iff \varphi = C_1\cosh\kappa x + C_2\sinh\kappa x$.

Proof. "⇐": differentiate (F1b). "⇒": (a) $u \equiv \varphi - \varphi(0)\cos kx - \frac{\varphi'(0)}{k}\sin kx$ solves $u'' = -k^2u$ with $u(0) = u'(0) = 0$. Then $\frac{\mathrm{d}}{\mathrm{d}x}(u'^2 + k^2u^2) = 2u'(u'' + k^2u) = 0$. So $u'^2 + k^2u^2$ is constant, equal to its value 0 at $x = 0$; hence $u \equiv 0$. (b) φ' is constant. (c) $u \equiv \varphi - \varphi(0)\cosh\kappa x - \frac{\varphi'(0)}{\kappa}\sinh\kappa x$ solves $u'' = \kappa^2 u$ with $u(0) = u'(0) = 0$. Let $v \equiv u' - \kappa u$. Then $v' = \kappa^2 u - \kappa u' = -\kappa v$, so $(e^{\kappa x}v)' = 0$ and $v \equiv v(0) = 0$. Then $(e^{-\kappa x}u)' = e^{-\kappa x}v = 0$, so $u \equiv u(0) = 0$. □

Each solution space is two-dimensional, spanned by one even and one odd function.

Own step. (standard) No page-checked reference. The "conserved combination" trick is the Wronskian argument of [Tong-QM, pp. 49–50].

F5 Integrating factor. If $y' + p(x)\,y = q(x)$ and $P' = p$, then $(e^P y)' = e^P(y' + py) = e^P q$ (product rule).

Own step. (standard) No page-checked reference.

F6 Gaussian integral. For $s > 0$: $\displaystyle\int_0^\infty e^{-su^2}\mathrm{d}u = \frac{1}{2}\sqrt{\frac{\pi}{s}}$ and $\displaystyle\int_{-\infty}^\infty e^{-su^2}\mathrm{d}u = \sqrt{\frac{\pi}{s}}$.

Proof (Calc 2, polar coordinates). Let $g \equiv \int_0^\infty e^{-u^2}\mathrm{d}u$. Then $g^2 = \int_0^\infty\!\int_0^\infty e^{-(u^2+v^2)}\mathrm{d}u\,\mathrm{d}v = \int_0^{\pi/2}\!\int_0^\infty e^{-r^2} r\,\mathrm{d}r\,\mathrm{d}\phi = \frac{\pi}{2}\cdot\frac{1}{2}$, so $g = \frac{\sqrt{\pi}}{2}$. Then substitute $u \to \sqrt{s}\,u$.

Source. Result: [Tong-SP, p. 35].

Own step. (standard) The proof.

F7 Error function. $\displaystyle\operatorname{erf} z = \frac{2}{\sqrt{\pi}}\int_0^z e^{-t^2}\mathrm{d}t, \qquad \operatorname{erfc} z = 1 - \operatorname{erf} z = \frac{2}{\sqrt{\pi}}\int_z^\infty e^{-t^2}\mathrm{d}t.$

(a) $\operatorname{erf} 0 = 0$, $\operatorname{erf}(\infty) = 1$ (by F6), and $\operatorname{erf}(-z) = -\operatorname{erf} z$.

(b) $(\operatorname{erf} z)' = \frac{2}{\sqrt{\pi}}e^{-z^2}$ (fundamental theorem of calculus). With the chain rule, $\displaystyle\frac{\mathrm{d}}{\mathrm{d}a}\operatorname{erf}\sqrt{a/2} = \frac{2}{\sqrt{\pi}}e^{-a/2}\cdot\frac{1}{2\sqrt{2a}} = \frac{e^{-a/2}}{\sqrt{2\pi a}}$ and $\displaystyle\frac{\mathrm{d}}{\mathrm{d}a}\operatorname{erf}\sqrt{a} = \frac{e^{-a}}{\sqrt{\pi a}}$.

(c) Small z: $\operatorname{erf} z = \frac{2}{\sqrt{\pi}}(z - \frac{z^3}{3} + \dots)$. Exactly, $\displaystyle e^{z^2}\operatorname{erf} z = \frac{2}{\sqrt{\pi}}\sum_{n\ge 0}\frac{2^n z^{2n+1}}{(2n+1)!!}$, where $(2n+1)!! \equiv 1\cdot 3\cdot 5\cdots(2n+1)$.

(d) Large z: $\displaystyle\operatorname{erfc} z = \frac{e^{-z^2}}{\sqrt{\pi}\,z}\Big(1 - \frac{1}{2z^2} + \dots\Big)$.

Source. [DLMF] 7.2.1–2 (definitions), 7.4.1 (odd), 7.6.1–2 (c), 7.12.1 (d).

F8 Differentiating under the integral sign. $\frac{\mathrm{d}}{\mathrm{d}s}\int f(s,u)\,\mathrm{d}u = \int \frac{\partial f}{\partial s}\,\mathrm{d}u$ holds when f and $\partial f/\partial s$ are continuous on a finite interval. On an infinite interval it needs, in addition, that $\int \partial_s f\,\mathrm{d}u$ converges uniformly, for example $|\partial_s f| \le g(u)$ with $\int g\,\mathrm{d}u < \infty$. In every use below, the integrand and its s-derivative are either continuous on a finite interval or bounded by $(1+u^2)e^{-s_0u^2}$ for $s \ge s_0 > 0$.

Source. [DLMF] 1.5.22 (finite interval, with the conditions above) and 1.5.23–24 (infinite interval).

F9 Integrating a series term by term. $\int_a^b \sum_n f_n = \sum_n \int_a^b f_n$ if the series converges uniformly on $[a,b]$. An example used below: $\sum_{\ell\ge 1}\frac{a^\ell v^{\ell-1}}{\ell!}$ on $0 \le v \le 1$. Its terms are bounded by $\frac{a^\ell}{\ell!}$, and $\sum_\ell \frac{a^\ell}{\ell!} < \infty$ (Weierstrass test).

Source. [DLMF] 1.9.68. (Weierstrass test: standard; no page-checked reference.)

F10 From a sum to an integral (Euler–Maclaurin, even functions). Let F be smooth and even on $\mathbb{R}$ ($F(-y) = F(y)$), with F and all its derivatives integrable and tending to 0 as $y \to \infty$. Then

$$\sum_{j=1}^\infty F(j) = \int_0^\infty F(y)\,\mathrm{d}y - \frac{1}{2}F(0) + R, \qquad |R| \le \frac{1}{12}\int_0^\infty |F''|\,\mathrm{d}y, \quad |R| \le \frac{4}{(2\pi)^{2m}}\int_0^\infty |F^{(2m)}|\,\mathrm{d}y \ (m \ge 1). \tag{1.1}$$

Step 1. On $[j, j+1]$ let $P_1(y) = y - j - \frac{1}{2}$, so $P_1' = 1$, $P_1(j) = -\frac{1}{2}$, $P_1(j+1) = \frac{1}{2}$. Integrate by parts:

$$\int_j^{j+1} P_1 F'\,\mathrm{d}y = \big[P_1 F\big]_j^{j+1} - \int_j^{j+1} F\,\mathrm{d}y = \frac{F(j)+F(j+1)}{2} - \int_j^{j+1} F\,\mathrm{d}y.$$

Sum over $j = 0, 1, 2, \dots$, using $\sum_{j\ge 0}\frac{F(j)+F(j+1)}{2} = \frac{F(0)}{2} + \sum_{j\ge 1} F(j)$. The result is (1.1) with $R = \int_0^\infty P_1 F'\,\mathrm{d}y$, where P_1 is the sawtooth $y - \lfloor y\rfloor - \frac{1}{2}$.

Step 2. Let $P_2(y) \equiv \frac{1}{2}(s^2 - s + \frac{1}{6})$ with $s = y - \lfloor y \rfloor$. It is continuous, has period 1, satisfies $P_2' = P_1$, and $|P_2| \le \frac{1}{12}$. Integrating by parts again, $R = \left[P_2 F'\right]_0^\infty - \int_0^\infty P_2 F''\,\mathrm{d}y = -\int_0^\infty P_2 F''\,\mathrm{d}y$, because $F'(0) = 0$ (F even, so F' is odd). Hence $|R| \le \frac{1}{12}\int |F''|$.

Step 3. Repeating this, every boundary term at $y = 0$ contains either an odd derivative $F^{(2i-1)}(0) = 0$ or a factor $P_{2i+1}(0) = 0$. The result is $R = -\int_0^\infty P_{2m} F^{(2m)}\,\mathrm{d}y$ with $\max|P_{2m}| = |B_{2m}|/(2m)! < 4/(2\pi)^{2m}$, where B_{2m} are the Bernoulli numbers. □

Example, exact. For $F(y) = e^{-\omega y^2}$ ($\omega > 0$), $\sum_{n=-\infty}^{\infty} e^{-\omega n^2} = \sqrt{\pi/\omega}\,(1 + 2\sum_{\ell\ge1} e^{-\pi^2\ell^2/\omega})$. With $\sum_{j\ge1} = \frac{1}{2}(\sum_{n=-\infty}^{\infty} - 1)$ and F6, this gives

$$R = \sqrt{\frac{\pi}{\omega}} \sum_{\ell\ge1} e^{-\pi^2\ell^2/\omega}. \tag{1.2}$$

So R is tiny once F is wide compared with the spacing 1 ($\omega \ll 1$) (C9a).

Source. General formula: [DLMF] 2.10.1. $P_n = \widetilde{B}_n/n!$ (periodic Bernoulli functions): [DLMF] 24.2.11–12. Bounds: [DLMF] 24.9.1, 24.9.8–9 (C9c). The Gaussian sum: [DLMF] 1.8.16 with $x = 0$.

Own step. (standard) Steps 1–2 and the specialisation to even F.

F11 A Gaussian–Lorentzian integral. For $s > 0$ and $c > 0$:

$$\mathcal{F}(s) \equiv \int_0^\infty \frac{e^{-su^2}}{u^2 + c^2}\,\mathrm{d}u = \frac{\pi}{2c}\, e^{sc^2}\,\mathrm{erfc}(c\sqrt{s}). \tag{1.3}$$

Proof:

$$\begin{aligned}
\mathcal{F}'(s) &= -\int_0^\infty \frac{u^2\, e^{-su^2}}{u^2 + c^2}\,\mathrm{d}u && \text{F8}\\
\mathcal{F}' - c^2\mathcal{F} &= -\int_0^\infty \frac{(u^2 + c^2)\, e^{-su^2}}{u^2 + c^2}\,\mathrm{d}u = -\frac{1}{2}\sqrt{\frac{\pi}{s}} && \text{F6}\\
(e^{-c^2 s}\mathcal{F})' &= -\frac{1}{2}\sqrt{\frac{\pi}{s}}\, e^{-c^2 s} && \text{F5 with } p = -c^2\\
e^{-c^2 s}\mathcal{F}(s) &= \frac{1}{2}\int_s^\infty \sqrt{\frac{\pi}{\sigma}}\, e^{-c^2\sigma}\mathrm{d}\sigma && \text{integrate from } s \text{ to } \infty;\ e^{-c^2 S}\mathcal{F}(S) \to 0 \text{ since } \mathcal{F} \le \tfrac{\pi}{2c} \text{ (F2a)}\\
&= \frac{\sqrt{\pi}}{c}\int_{c\sqrt{s}}^\infty e^{-v^2}\mathrm{d}v = \frac{\pi}{2c}\,\mathrm{erfc}\,(c\sqrt{s}) && \sigma = v^2/c^2,\ \mathrm{d}\sigma = 2v\,\mathrm{d}v/c^2;\ \text{F7}
\end{aligned}$$

Multiply by $e^{c^2 s}$. As a check, $s \to 0$ gives $\frac{\pi}{2c}$, as F2a requires (C10).

Source. Result: [DLMF] 7.7.1 (put $u = ct$, $z = c\sqrt{s}$).

Own step. The proof. No reference found for this route.

F12 A polynomial integral. For $n = 0, 1, 2, \ldots$: $W_n \equiv \int_{-1}^{1} (1 - t^2)^n\,\mathrm{d}t = \dfrac{2^{n+1}\, n!}{(2n+1)!!}$.

Proof: $W_0 = 2$. For $n \ge 1$, integrate by parts with $\mathrm{d}v = \mathrm{d}t$: $W_n = \left[t(1-t^2)^n\right]_{-1}^{1} + 2n\int_{-1}^{1} t^2(1-t^2)^{n-1}\mathrm{d}t = 2n\int_{-1}^{1}\left[1 - (1-t^2)\right](1-t^2)^{n-1}\mathrm{d}t = 2n\,(W_{n-1} - W_n)$. So $W_n = \frac{2n}{2n+1}W_{n-1} = 2\cdot\frac{2\cdot4\cdots2n}{3\cdot5\cdots(2n+1)} = \frac{2\cdot2^n\, n!}{(2n+1)!!}$. □ (C10)

Source. Equivalent to Euler's Beta integral, [DLMF] 5.12.1 (with 5.4.6 and 5.5.1).

Own step. (standard) The recursion.

F13–F15 are used only in the second route of §4.7.

F13 Complex numbers. $i^2 = -1$ and $e^{i\theta} = \cos\theta + i\sin\theta$, so $e^{\pm ikx} = \cos kx \pm i\sin kx$. If $w = u + iv$ (with u, v real) solves a linear ODE whose coefficients are real, then u and v each solve it: take the real and imaginary parts of the equation.

Source. $\cos z \pm i\sin z = e^{\pm iz}$: [DLMF] 4.14.3.

Own step. (standard) The real/imaginary-part remark.

F14 Gamma function. $\Gamma(z) = \int_0^\infty e^{-t}t^{z-1}\mathrm{d}t$ for $\operatorname{Re} z > 0$. Integration by parts gives $\Gamma(z+1) = \left[-e^{-t}t^z\right]_0^\infty + z\int_0^\infty e^{-t}t^{z-1}\mathrm{d}t = z\,\Gamma(z)$; with $\Gamma(1) = 1$ this gives $\Gamma(n+1) = n!$. Γ extends to all complex z except $z = 0, -1, -2, \ldots$, where it has poles. It has no zeros, so $1/\Gamma(z)$ is finite everywhere and vanishes exactly at $z = 0, -1, -2, \ldots$

Source. [DLMF] 5.2.1 (definition), 5.5.1 ($\Gamma(z+1) = z\Gamma(z)$); poles, no zeros and $1/\Gamma$: the text after 5.2.1 in §5.2(i).

Own step. (standard) The integration by parts.

F15 Hypergeometric series. With the rising factorial $(a)_0 = 1$, $(a)_s = a(a+1)\cdots(a+s-1)$, $F(a,b;c;z) = \sum_{s\geq 0} \dfrac{(a)_s(b)_s}{(c)_s\, s!}\, z^s = 1 + \dfrac{ab}{c}z + \dfrac{a(a+1)b(b+1)}{c(c+1)\,2!}z^2 + \ldots$ If $b = -m$ with $m = 0, 1, 2, \ldots$, then $(b)_s = 0$ for $s > m$ and the series stops: F is a polynomial of degree m. For $m = 1$: $F(a, -1; c; z) = 1 - \frac{a}{c}\,z$.

Source. [DLMF] 5.2.4–5 (rising factorial), 15.2.1 (series), 15.2.4 (terminating case).

2 Physics toolkit

P1 Quantum energy levels. $H = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x)$ is a linear map (F3). The allowed energies are its eigenvalues: the values of E for which $H\psi = E\psi$ has a solution $\psi \not\equiv 0$ obeying the boundary conditions. ψ and $C\psi$ ($C \neq 0$) are the same state. At a hard wall $V = +\infty$, which forces $\psi = 0$ there; ψ is continuous, so $\psi(\pm L/2) = 0$.

Source. [Tong-QM, p. 9, Eq. (1.1); p. 10, Eq. (1.2); p. 68]. Hard walls: [Tong-QM, pp. 24–25].

Example: the empty box ($V = 0$ for $|x| < L/2$), solved with F4. Section 4 repeats exactly these steps for the well.

- $E = \frac{\hbar^2k^2}{2m} > 0$. Then $\psi = C_1\cos kx + C_2\sin kx$, and $\psi(\frac{L}{2}) \pm \psi(-\frac{L}{2}) = 0$ means $C_1\cos\frac{kL}{2} = 0$ and $C_2\sin\frac{kL}{2} = 0$. A nonzero ψ needs $\cos\frac{kL}{2} = 0$ or $\sin\frac{kL}{2} = 0$, i.e. $kL = n\pi$ with $n = 1, 2, \ldots$
- $E = 0$: $C_1 + C_2x$ vanishes at $\pm\frac{L}{2}$ only if $C_1 = C_2 = 0$. $E < 0$: $C_1\cosh\frac{\kappa L}{2} = 0 = C_2\sinh\frac{\kappa L}{2}$ forces $C_1 = C_2 = 0$.

$$E_n = n^2E_1, \qquad E_1 = \frac{\pi^2\hbar^2}{2mL^2}, \qquad n = 1, 2, 3, \ldots \tag{2.1}$$

Source. [Tong-QM, p. 25, Eq. (2.10)] (box $0 < x < L$, same levels).

Own step. (standard) The derivation via F4.

P2 Parity. If $V(-x) = V(x)$ and $\psi(x)$ solves $H\psi = E\psi$, then so does $\psi(-x)$ (chain rule). The solutions can therefore be chosen even or odd. If $\psi = c_e\psi_e + c_o\psi_o$ with ψ_e even and ψ_o odd, then $\psi(\pm\frac{L}{2}) = c_e\psi_e(\frac{L}{2}) \pm c_o\psi_o(\frac{L}{2})$. Adding and subtracting,

$$\psi(\pm L/2) = 0 \iff c_e\,\psi_e(L/2) = 0 \ \text{ and } \ c_o\,\psi_o(L/2) = 0. \tag{2.2}$$

Source. Parity: [Tong-Topics, p. 196].

Own step. (standard) Eq. (2.2).

P3 Canonical ensemble. In equilibrium with a bath at temperature T, level n is occupied with probability $p_n = e^{-\beta E_n}/Z$, where $Z = \sum_n e^{-\beta E_n}$ and $\beta = 1/k_B T$.

Source. [Tong-SP, p. 18, Eqs. (1.19)–(1.22)].

P4 Heat capacity from Z.

$$U \equiv \langle E \rangle = -\frac{\partial \ln Z}{\partial \beta}, \quad \frac{\partial^2 \ln Z}{\partial \beta^2} = \langle E^2 \rangle - \langle E \rangle^2 \equiv \mathrm{Var}(E), \quad C_v \equiv \frac{\mathrm{d}U}{\mathrm{d}T} = k_B \beta^2 \frac{\partial^2 \ln Z}{\partial \beta^2} = k_B \beta^2 \,\mathrm{Var}(E). \tag{2.3}$$

Proof: $\partial_\beta Z = -\sum_n E_n e^{-\beta E_n}$, so $-\partial_\beta \ln Z = \sum_n E_n p_n$. Next $\partial_\beta^2 Z = \sum_n E_n^2 e^{-\beta E_n}$, so $\partial_\beta^2 \ln Z = \frac{\partial_\beta^2 Z}{Z} - \left(\frac{\partial_\beta Z}{Z}\right)^2 = \langle E^2 \rangle - \langle E \rangle^2$. Finally $\frac{\mathrm{d}\beta}{\mathrm{d}T} = -\frac{1}{k_B T^2} = -k_B \beta^2$ and $\frac{\mathrm{d}U}{\mathrm{d}T} = \frac{\mathrm{d}U}{\mathrm{d}\beta}\frac{\mathrm{d}\beta}{\mathrm{d}T}$. □

Source. [Tong-SP, p. 20, Eqs. (1.25)–(1.27)].

Consequences:

(a) $C_v \geq 0$.

(b) Multiplying Z by a constant, or shifting every E_n by the same constant, leaves C_v unchanged: $\ln Z$ changes by a constant or by a term linear in β.

(c) If $Z = Z_1 Z_2$, then $C_v = C_1 + C_2$.

(d) If $a = D\beta$ with $D > 0$ constant, then $\beta \partial_\beta = a \partial_a$ and $\beta^2 \partial_\beta^2 = a^2 \partial_a^2$, so

$$\frac{C_v}{k_B} = a^2 \frac{\mathrm{d}^2 \ln Z}{\mathrm{d}a^2} = a^2 \left[\frac{Z''}{Z} - \left(\frac{Z'}{Z}\right)^2 \right], \qquad ' = \frac{\mathrm{d}}{\mathrm{d}a}. \tag{2.4}$$

(e) *Sign rule.* If $\ln Z_1 - \ln Z_2 = -C\beta^p$ with $C > 0$, then $(C_1 - C_2)/k_B = -C\,p(p-1)\,\beta^p$, which is positive iff $0 < p < 1$ (C18a).

(f) *Two-level (Schottky) model.* Take one ground state and g states at height Δ, so $Z = 1 + g e^{-\beta\Delta}$. Let $P = \frac{g e^{-\beta\Delta}}{1 + g e^{-\beta\Delta}}$, the probability of being up. Then $-\partial_\beta \ln Z = \Delta P$ and $\partial_\beta P = -\Delta P(1-P)$, so $C_v/k_B = (\beta\Delta)^2 P(1-P)$. The peak is about $(\beta\Delta)^2/4$, at $P = \frac{1}{2}$, i.e. $g e^{-\beta\Delta} = 1$. At low T, $C_v \approx k_B (\beta\Delta)^2 g\, e^{-\beta\Delta}$, which goes to 0 exponentially because of the gap Δ.

Source. (f): the two-state system and its exponential fall-off due to the gap, [Tong-SP, p. 13, Eq. (1.14)].

Own step. (a)–(e), and (f) with g upper states.

P5 Classical particle.

$$Z_{\mathrm{cl}} = \frac{1}{2\pi\hbar} \int \mathrm{d}x \int_{-\infty}^{\infty} \mathrm{d}p\; e^{-\beta\left(\frac{p^2}{2m} + V(x)\right)} = \frac{1}{\lambda_{\mathrm{th}}} \int e^{-\beta V(x)}\,\mathrm{d}x, \qquad \lambda_{\mathrm{th}} \equiv \sqrt{\frac{2\pi\hbar^2\beta}{m}}. \tag{2.5}$$

The second form uses $\int e^{-\beta p^2/2m} \mathrm{d}p = \sqrt{2\pi m/\beta}$ (F6 with $s = \beta/2m$) and $\frac{1}{2\pi\hbar}\sqrt{\frac{2\pi m}{\beta}} = \sqrt{\frac{m}{2\pi\hbar^2\beta}} = \frac{1}{\lambda_{\mathrm{th}}}$. The factor $\frac{1}{2\pi\hbar}$ makes Z dimensionless and makes it agree with the quantum Z at high T; by P4(b) it does not affect C_v. λ_{th} is the thermal de Broglie wavelength, and L/λ_{th} counts how many of them fit in a length L.

Source. [Tong-SP, p. 32, Eq. (2.1) and the $1/h$ factor; pp. 33–34, the one-dimensional form derived from the quantum Z; p. 35, Eqs. (2.5)–(2.6)].

Split $Z_{\mathrm{cl}} = \left[\sqrt{m/2\pi\hbar^2}\;\beta^{-1/2}\right] \times \int e^{-\beta V}\mathrm{d}x$ and use P4(c). The first factor gives $\beta^2\partial_\beta^2(-\frac{1}{2}\ln\beta) = \frac{1}{2}$. The second factor is P4 with "levels" $V(x)$ labelled by x. Together,

$$C_v^{\mathrm{cl}} = \frac{k_B}{2} + k_B \beta^2 \,\mathrm{Var}_x(V), \qquad \text{averaged over } P(x) = \frac{e^{-\beta V(x)}}{\int e^{-\beta V}\,\mathrm{d}x}. \tag{2.6}$$

Equipartition: each quadratic term in H contributes $\frac{1}{2}k_BT$ to U. A classical harmonic oscillator therefore has $U = k_BT$ and $C_v = k_B$.

Source. Equipartition: [Tong-SP, p. 37]. Oscillator: [Tong-SP, p. 47, Eq. (2.20)] and [LGK, p. 95].

Own step. (standard) Eq. (2.6).

3 The model

$$V(x) = \begin{cases} -\dfrac{\hbar^2}{2m}\,\alpha^2\,\dfrac{\lambda(\lambda-1)}{\cosh^2(\alpha x)}, & |x| < L/2, \\ +\infty, & |x| \geq L/2. \end{cases} \tag{3.1}$$

Source. Inside the box, the "modified Pöschl–Teller" row of [DJdC, p. 021013-9, Table I], where q stands for our x. The table takes it from Flügge, which is not checked here (no free copy found).

For $\lambda = 2$, $\lambda(\lambda-1) = 2$ and

$$V(x) = -\frac{\hbar^2\alpha^2}{m}\,\frac{1}{\cosh^2(\alpha x)} \qquad (|x| < L/2). \tag{3.2}$$

The depth is $\hbar^2\alpha^2/m$, at $x = 0$. By F1e, $|V(x)| < \frac{4\hbar^2\alpha^2}{m}e^{-2\alpha|x|}$, so the width is about $1/\alpha$ and $|V(\pm L/2)| < \frac{4\hbar^2\alpha^2}{m}e^{-\alpha L}$.

Why $\lambda = 2$. The well $-n(n+1)\,\mathrm{sech}^2 x$ "holds n bound states" and is reflectionless [CKS, p. 77, Eq. (312); pp. 41–42]. CKS write sech $= 1/\cosh$ and use units $\hbar = 2m = \alpha = 1$. Since $n(n+1) = \lambda(\lambda-1) = 2$ gives $n = 1$, $\lambda = 2$ is the smallest integer λ that binds, and it binds exactly one state; §4 proves this inside the box.

Only two dimensionless numbers. The inputs are $\hbar$, m, α (an inverse length), L and k_BT. From $\hbar$, m and α one can form exactly one length, $1/\alpha$ (the well width), and one energy, $\hbar^2\alpha^2/m$ (the depth). A dimensionless output such as C_v/k_B can therefore depend only on

$$\alpha L \;\; \text{(box length in well widths)} \qquad \text{and} \qquad a \equiv \frac{\beta\hbar^2\alpha^2}{m} = \frac{\text{well depth}}{k_BT}. \tag{3.3}$$

With λ_{th} from (2.5), $\lambda_{\mathrm{th}}^2 = \frac{2\pi\hbar^2\beta}{m} = \frac{2\pi a}{\alpha^2}$, hence

$$\alpha\lambda_{\mathrm{th}} = \sqrt{2\pi a}, \qquad \frac{L}{\lambda_{\mathrm{th}}} = \frac{\alpha L}{\sqrt{2\pi a}}, \qquad \frac{k_BT}{\hbar^2\alpha^2/2m} = \frac{2}{a}. \tag{3.4}$$

Own step. The dimensional argument and Eq. (3.4).

4 The energy levels

Inside the box we must solve

$$H_{\mathrm{PT}}\psi \equiv -\frac{\hbar^2}{2m}\psi'' - \frac{\hbar^2\alpha^2}{m}\frac{1}{\cosh^2(\alpha x)}\,\psi = E\,\psi \quad (|x| < L/2), \qquad \psi(\pm L/2) = 0. \tag{4.1}$$

The plan has three steps. (i) Factor H_{PT}. (ii) Map every solution of the free equation to a solution of (4.1), and show that these are all of its solutions. (iii) Impose $\psi(\pm L/2) = 0$. §4.7 reaches the same solutions by a second route (associated Legendre functions), and §4.8 compares the two.

4.1 Factorization

Define

$$W(x) \equiv \frac{\hbar\alpha}{\sqrt{2m}}\tanh(\alpha x), \qquad A \equiv \frac{\hbar}{\sqrt{2m}}\frac{\mathrm{d}}{\mathrm{d}x} + W, \qquad A^\dagger \equiv -\frac{\hbar}{\sqrt{2m}}\frac{\mathrm{d}}{\mathrm{d}x} + W. \tag{4.2}$$

(Here "†" is only a name.) For any twice-differentiable f, by the product rule:

$$\begin{aligned} A^\dagger A f &= \Big(-\tfrac{\hbar}{\sqrt{2m}}\frac{\mathrm{d}}{\mathrm{d}x} + W\Big)\Big(\tfrac{\hbar}{\sqrt{2m}}f' + Wf\Big) = -\frac{\hbar^2}{2m}f'' - \tfrac{\hbar}{\sqrt{2m}}(W'f + Wf') + \tfrac{\hbar}{\sqrt{2m}}Wf' + W^2 f \\ &= -\frac{\hbar^2}{2m}f'' + \Big(W^2 - \tfrac{\hbar}{\sqrt{2m}}W'\Big)f, \end{aligned} \tag{4.3}$$

$$AA^\dagger f = \tfrac{\hbar}{\sqrt{2m}}\Big(-\tfrac{\hbar}{\sqrt{2m}}f' + Wf\Big)' + W\Big(-\tfrac{\hbar}{\sqrt{2m}}f' + Wf\Big) = -\frac{\hbar^2}{2m}f'' + \Big(W^2 + \tfrac{\hbar}{\sqrt{2m}}W'\Big)f. \tag{4.4}$$

Now $W^2 = \frac{\hbar^2\alpha^2}{2m}\tanh^2(\alpha x)$ and $\frac{\hbar}{\sqrt{2m}}W' = \frac{\hbar^2\alpha^2}{2m}\frac{1}{\cosh^2(\alpha x)}$ (F1b). With $\tanh^2 = 1 - 1/\cosh^2$ (F1a):

$$W^2 - \tfrac{\hbar}{\sqrt{2m}}W' = \frac{\hbar^2\alpha^2}{2m}\Big(1 - \frac{2}{\cosh^2(\alpha x)}\Big) = \frac{\hbar^2\alpha^2}{2m} - \frac{\hbar^2\alpha^2}{m}\frac{1}{\cosh^2(\alpha x)}, \qquad W^2 + \tfrac{\hbar}{\sqrt{2m}}W' = \frac{\hbar^2\alpha^2}{2m}.$$

Therefore, with $H_0 \equiv -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}$ the free particle,

$$A^\dagger A = H_{\mathrm{PT}} + \frac{\hbar^2\alpha^2}{2m}, \qquad AA^\dagger = H_0 + \frac{\hbar^2\alpha^2}{2m}. \tag{4.5}$$

(C2)

Source. [CKS, p. 13, Eqs. (3)–(4); p. 14, Eqs. (5), (7); p. 18, Eqs. (32)–(33)]. CKS take $W = A\tanh x$; with $A = \hbar/\sqrt{2m}$ their partner V_2 is constant. Ours is theirs with $x \to \alpha x$ and $A \to \hbar\alpha/\sqrt{2m}$.

4.2 Intertwining

$$H_{\mathrm{PT}}A^\dagger = A^\dagger H_0, \qquad A\,H_{\mathrm{PT}} = H_0 A. \tag{4.6}$$

Proof (associativity, F3): $H_{\mathrm{PT}}A^\dagger = (A^\dagger A - \frac{\hbar^2\alpha^2}{2m})A^\dagger = A^\dagger(AA^\dagger) - \frac{\hbar^2\alpha^2}{2m}A^\dagger = A^\dagger(H_0 + \frac{\hbar^2\alpha^2}{2m}) - \frac{\hbar^2\alpha^2}{2m}A^\dagger = A^\dagger H_0$. The second relation follows the same way. □ (C3) By the lemma of F3:

$$H_0\varphi = E\varphi \implies H_{\mathrm{PT}}(A^\dagger\varphi) = E\,(A^\dagger\varphi). \tag{4.7}$$

Source. [CKS, p. 14, Eqs. (8)–(11)].

4.3 All solutions at a given energy

Claim. Let $E \neq -\frac{\hbar^2\alpha^2}{2m}$. Then ψ solves $H_{\mathrm{PT}}\psi = E\psi$ if and only if $\psi = A^\dagger\varphi$ for some φ with $H_0\varphi = E\varphi$.

Proof. "If" is (4.7). "Only if": set $\varphi \equiv A\psi/(E + \frac{\hbar^2\alpha^2}{2m})$. By (4.6), $H_0\varphi = AH_{\mathrm{PT}}\psi/(\dots) = E\,\varphi$. By (4.5), $A^\dagger\varphi = A^\dagger A\psi/(\dots) = (H_{\mathrm{PT}} + \frac{\hbar^2\alpha^2}{2m})\psi/(\dots) = \psi$. □

So the solutions of the well equation are exactly $A^\dagger$ applied to the free solutions, which F4 lists completely.

Own step. CKS [p. 14, Eqs. (13)–(14)] map normalizable eigenfunctions this way. The "all solutions" statement, which the box needs, is Claude's.

4.4 The solutions, explicitly

Write $A^\dagger = \frac{\hbar}{\sqrt{2m}}\big(-\frac{\mathrm{d}}{\mathrm{d}x} + \alpha\tanh(\alpha x)\big)$ and apply it to the bases of F4.

(a) $E = \frac{\hbar^2k^2}{2m} > 0$, $k > 0$: $A^\dagger \cos kx = \frac{\hbar}{\sqrt{2m}}\,\psi_o$ and $A^\dagger \sin kx = -\frac{\hbar}{\sqrt{2m}}\,\psi_e$, where

$$\psi_e(x) = k\cos kx - \alpha\tanh(\alpha x)\sin kx \ \ \text{(even)}, \qquad \psi_o(x) = \alpha\tanh(\alpha x)\cos kx + k\sin kx \ \ \text{(odd)}. \tag{4.8}$$

(b) $E = 0$: $A^\dagger 1 = \frac{\hbar}{\sqrt{2m}}\,\alpha\tanh(\alpha x)$ (odd) and $A^\dagger x = \frac{\hbar}{\sqrt{2m}}\big(\alpha x\tanh(\alpha x) - 1\big)$ (even).

(c) $E = -\frac{\hbar^2\kappa^2}{2m} < 0$, $\kappa > 0$, $\kappa \neq \alpha$: $A^\dagger \cosh\kappa x = \frac{\hbar}{\sqrt{2m}}\,\chi_o$ and $A^\dagger \sinh\kappa x = -\frac{\hbar}{\sqrt{2m}}\,\chi_e$, where

$$\chi_e(x) = \kappa\cosh\kappa x - \alpha\tanh(\alpha x)\sinh\kappa x \ \ \text{(even)}, \qquad \chi_o(x) = \alpha\tanh(\alpha x)\cosh\kappa x - \kappa\sinh\kappa x \ \ \text{(odd)}. \tag{4.9}$$

The parities follow from F1c. None of these functions is identically zero: $\psi_e(0) = k$, $\psi_o'(0) = \alpha^2 + k^2$, $\chi_e(0) = \kappa$, $\chi_o'(0) = \alpha^2 - \kappa^2 \neq 0$. So for $E \neq -\frac{\hbar^2\alpha^2}{2m}$ the general solution is $c_e(\text{even}) + c_o(\text{odd})$ (C4).

(d) $E = -\frac{\hbar^2\alpha^2}{2m}$ ($\kappa = \alpha$). Here $A^\dagger\cosh\alpha x = 0$ (by F1f), so the claim above does not apply; solve directly instead. Let $u \equiv A\psi$. Then $A^\dagger u = A^\dagger A\psi = (H_{\mathrm{PT}} + \frac{\hbar^2\alpha^2}{2m})\psi = 0$, i.e. $u' = \alpha\tanh(\alpha x)\,u$, so $\big(u/\cosh\alpha x\big)' = 0$ (F5) and $u = C\cosh\alpha x$. Next, $A\psi = C\cosh\alpha x$ reads $\psi' + \alpha\tanh(\alpha x)\psi = C'\cosh\alpha x$ with $C' = C\sqrt{2m}/\hbar$. So $(\psi\cosh\alpha x)' = C'\cosh^2\alpha x$ (F5), and integrating with F1d, $\psi\cosh\alpha x = C'\big(\frac{x}{2} + \frac{\sinh 2\alpha x}{4\alpha}\big) + C''$. Hence

$$\psi = C''y_1 + C'y_2, \qquad y_1 = \frac{1}{\cosh(\alpha x)} \ \ \text{(even)}, \qquad y_2 = \frac{x/2 + \sinh(2\alpha x)/4\alpha}{\cosh(\alpha x)} \ \ \text{(odd)}. \tag{4.10}$$

Note that $Ay_1 = 0$: y_1 is the "zero mode" of A (C5).

Source. Zero mode $A\psi_0 = 0$ and its role: [CKS, p. 14, Eq. (6); p. 15].

Own step. (a)–(d) as complete solution sets, and (d) itself.

4.5 Imposing the walls

By (2.2), E is a level if and only if the even or the odd solution vanishes at $x = L/2$. There $\alpha\tanh(\alpha x) = \alpha\tanh(\alpha L/2)$. We use $\alpha L\tanh(\alpha L/2) > 2$, which holds for $\alpha L > 2.40$ (C6).

(i) $E > 0$. Even: $\psi_e(\frac{L}{2}) = 0 \iff k\cos\frac{kL}{2} = \alpha\tanh(\alpha L/2)\sin\frac{kL}{2} \iff \tan\frac{kL}{2} = \frac{k}{\alpha\tanh(\alpha L/2)}$. (If $\cos\frac{kL}{2} = 0$, the left side vanishes but the right side does not, since then $\sin\frac{kL}{2} = \pm 1$.) Odd: $\psi_o(\frac{L}{2}) = 0 \iff \alpha\tanh(\alpha L/2)\cos\frac{kL}{2} + k\sin\frac{kL}{2} = 0 \iff \tan\frac{kL}{2} = -\frac{\alpha\tanh(\alpha L/2)}{k}$. Let $\varphi \equiv \arctan\frac{k}{\alpha\tanh(\alpha L/2)} \in (0, \frac{\pi}{2})$. Then the even condition is $\tan\frac{kL}{2} = \tan\varphi$, and the odd one is $\tan\frac{kL}{2} = -\frac{1}{\tan\varphi} = \tan(\varphi + \frac{\pi}{2})$ (F2d). By F2c, even states have $\frac{kL}{2} = \varphi + n\pi$ and odd states $\frac{kL}{2} = \varphi + \frac{\pi}{2} + n\pi$. Both fit in one line:

$$k_jL - 2\arctan\frac{k_j}{\alpha\tanh(\alpha L/2)} = j\pi, \qquad E_j = \frac{\hbar^2k_j^2}{2m}, \tag{4.11}$$

with even j for even states and odd j for odd states.

Exactly one root for each $j \geq 1$, and none for $j \leq 0$. Let $f(k) \equiv kL - 2\arctan\frac{k}{\alpha\tanh(\alpha L/2)}$. Then $f(0) = 0$ and, by F2a, $f'(k) = L - \frac{2\alpha\tanh(\alpha L/2)}{k^2 + \alpha^2\tanh^2(\alpha L/2)} \geq L - \frac{2}{\alpha\tanh(\alpha L/2)} > 0$. So f increases from 0 to ∞: $j \leq 0$ has no root with $k > 0$, and each $j \geq 1$ has exactly one, with $j\pi < k_jL < (j+1)\pi$ because $0 < 2\arctan < \pi$.

(ii) $E = 0$**.** At $x = \frac{L}{2}$ the odd solution equals $\frac{\hbar}{\sqrt{2m}}\alpha\tanh(\alpha L/2) \neq 0$, and the even one equals $\frac{\hbar}{\sqrt{2m}}(\frac{\alpha L}{2}\tanh(\alpha L/2) - 1) \neq 0$. So $E = 0$ is not a level.

(iii) $E < 0$**,** $\kappa \neq \alpha$**.** Even: $\chi_e(\frac{L}{2}) = 0 \iff \kappa\cosh\frac{\kappa L}{2} = \alpha\tanh(\alpha L/2)\sinh\frac{\kappa L}{2}$, i.e.

$$\kappa = \alpha\tanh(\alpha L/2)\,\tanh(\kappa L/2). \tag{4.12}$$

Let $g(\kappa) \equiv \alpha\tanh(\alpha L/2)\tanh\frac{\kappa L}{2} - \kappa$. Then $g(0) = 0$ and $g'(0) = \frac{\alpha L}{2}\tanh(\alpha L/2) - 1 > 0$. Also $g''(\kappa) = -\alpha\tanh(\alpha L/2)\frac{L^2}{2}\frac{\tanh(\kappa L/2)}{\cosh^2(\kappa L/2)} < 0$ for $\kappa > 0$, and $g(\kappa) < \alpha - \kappa$. So g' decreases strictly: g first rises, then falls strictly to $-\infty$. Hence (4.12) has exactly one root $\kappa_b > 0$, and $\kappa_b = \alpha\tanh(\alpha L/2)\tanh\frac{\kappa_b L}{2} < \alpha$; in particular $\kappa_b \neq \alpha$. Odd: $\chi_o(\frac{L}{2}) = 0 \iff \kappa\tanh\frac{\kappa L}{2} = \alpha\tanh(\alpha L/2)$. The left side increases strictly with κ, so the only root is $\kappa = \alpha$, which this case excludes. There is no odd level.

(iv) $E = -\frac{\hbar^2\alpha^2}{2m}$**.** From (4.10), $y_1(\frac{L}{2}) = 1/\cosh\frac{\alpha L}{2} > 0$ and $y_2(\frac{L}{2}) > 0$ (both terms positive). By (2.2) only $\psi \equiv 0$ is allowed, so this energy is not a level.

Spectrum. Exactly one negative level, the bound state $E_b = -\frac{\hbar^2\kappa_b^2}{2m}$ with κ_b from (4.12). Above it, the box levels E_j, $j = 1, 2, 3, \ldots$, from (4.11). These are exact for any $\alpha L > 2.40$.

Size of E_b. By F1e, $\alpha\tanh(\alpha L/2) = \alpha(1 - 2e^{-\alpha L} + \ldots)$, and $\tanh\frac{\kappa_b L}{2} = 1 - 2e^{-\kappa_b L} + \cdots = 1 - 2e^{-\alpha L} + \ldots$. Hence

$$\kappa_b = \alpha(1 - 4e^{-\alpha L} + \ldots), \qquad E_b = -\frac{\hbar^2\alpha^2}{2m}(1 - 8e^{-\alpha L} + \ldots), \tag{4.13}$$

where "…" is of order $\alpha L\, e^{-2\alpha L}$. At $\alpha L = 20$, $E_b/(\hbar^2\alpha^2/2m) = -0.99999998351077$, which equals $-(1 - 8e^{-20})$ to 15 digits (C7). The walls squeeze the bound state slightly, because the open-line state $1/\cosh(\alpha x)$ does not vanish at $\pm L/2$.

Own step. §4.5 (i)–(iv) and Eqs. (4.11)–(4.13). No reference found for the Pöschl–Teller well inside a box.

Source. Open line ($L \to \infty$): $E_b \to -\hbar^2\alpha^2/2m$ with $\psi_b \propto 1/\cosh(\alpha x)$ [CKS, p. 78], the case $n = 1$: $V_1 = -2\,\mathrm{sech}^2 x$, $E_1 = -1$, $\psi_1 \propto \mathrm{sech}\, x$ in units $\hbar = 2m = \alpha = 1$. The partner has "one fewer bound state" [CKS, p. 17], and the spectra agree apart from the zero mode [CKS, p. 15, Eqs. (12)–(14)]; the partner here is the free particle, which has none.

Table 1: Levels at $\alpha L = 20$, in units of $\hbar^2\alpha^2/2m$ (C6, C7).

	bound	$j = 1$	$j = 2$	$j = 3$	$j = 4$	$j = 5$
well + box, (4.11)–(4.12)	−0.99999998	0.030394	0.120836	0.269487	0.474202	0.733090
empty box, $n = j$: $(j\pi/\alpha L)^2$	–	0.024674	0.098696	0.222066	0.394784	0.616850

4.6 Physical picture: a phase shift, a level slide, one state removed

Far from the well, where $|x| \gg 1/\alpha$, F1e gives $\tanh(\alpha x) \to \pm 1$. Define the phase $\delta(k) \equiv \arctan\frac{\alpha}{k} \in (0, \frac{\pi}{2})$, so that $\cos\delta = \frac{k}{\sqrt{k^2+\alpha^2}}$ and $\sin\delta = \frac{\alpha}{\sqrt{k^2+\alpha^2}}$. For $x > 0$, (4.8) becomes

$$\begin{aligned} \psi_e &\to k\cos kx - \alpha\sin kx = \sqrt{k^2+\alpha^2}\,\cos\big(kx + \delta(k)\big), \\ \psi_o &\to \alpha\cos kx + k\sin kx = \sqrt{k^2+\alpha^2}\,\sin\big(kx + \delta(k)\big), \end{aligned} \tag{4.14}$$

with error at most $2\alpha e^{-2\alpha x}$ (C8). Outside the well, then, both parities are the free waves $\cos kx$ and $\sin kx$, shifted by the *same* phase $\delta(k)$ (Fig. 1a).

Source. Phase shifts of a symmetric potential, one per parity: [Tong-Topics, p. 199], where $S_{++} = t + r$ and $S_{--} = t - r$ are the phases $e^{2i\delta_\pm}$. Here $\delta_+ = \delta_- = \delta$, so the reflection amplitude $r = 0$. That the $\lambda = 2$ well is reflectionless: [CKS, p. 18; pp. 41–42, Eqs. (150)–(153)], with transmission $T_1 = \frac{1-ik}{-1-ik}$, $|T_1| = 1$, for $n = 1$.

The box in this language. Imposing $\psi(\frac{L}{2}) = 0$ on (4.14) gives $\cos(\frac{kL}{2} + \delta) = 0$ (even) or $\sin(\frac{kL}{2} + \delta) = 0$ (odd), i.e.

$$kL + 2\delta(k) = m\pi, \qquad m = 1, 2, \ldots, \tag{4.15}$$

to be compared with $kL = m\pi$ for the empty box (2.1). Since $2\delta = \pi - 2\arctan\frac{k}{\alpha}$ (F2b), this is (4.11) with $j = m - 1$ (and $\alpha \tanh(\alpha L/2) \to \alpha$).

Intuition

- The attractive well uses up the phase $2\delta(k)$. Each level therefore *slides down* from its empty-box value $(j+1)\pi/L$ by the fraction $2\delta(k)/\pi \in (0, 1)$ of one spacing: almost a full spacing for $k \ll \alpha$, almost none for $k \gg \alpha$ (Fig. 1b).
- The lowest empty-box level ($m = 1$) has no level below it to slide into. It leaves the continuum and becomes the bound state E_b.
- Counting states per unit k, from (4.11) with $\alpha \tanh(\alpha L/2) \to \alpha$:

$$\frac{\mathrm{d}j}{\mathrm{d}k} = \frac{L}{\pi} - \frac{1}{\pi}\frac{2\alpha}{k^2 + \alpha^2}. \tag{4.16}$$

  The first term is the empty box. The second is a Lorentzian of width α in k, the states the well removes, and its total is $\int_0^\infty \frac{1}{\pi}\frac{2\alpha}{k^2+\alpha^2}\mathrm{d}k = 1$ (F2a): exactly one state, the one that became E_b (C8) (Fig. 1c).

This is the one-dimensional form of Levinson's theorem (phase shift at $k = 0$ minus phase shift at $k = \infty$ equals $\pi\times$ the number of bound states). The name is given for orientation only; no page-checked source was found.

Own step. Eqs. (4.14)–(4.16) and their reading.

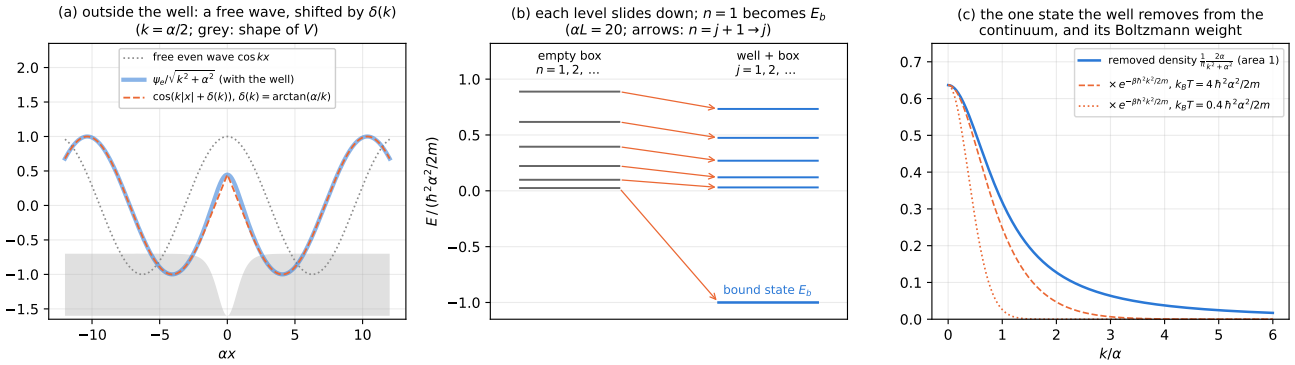


Figure 1: (a) The even solution (4.8) (blue) agrees outside the well with the free wave shifted by $\delta(k)$ (orange, dashed), (4.14). (b) Levels at $\alpha L = 20$: each empty-box level $n = j + 1$ slides down to j, and $n = 1$ becomes the bound state. (c) The removed-state density of (4.16) (area 1), and its Boltzmann-weighted versions at two temperatures, as they enter Z_q in §5.

4.7 A second route: associated Legendre functions

This is the route of most textbooks: a change of variable turns (4.1) into Legendre's equation, whose solutions are tabulated special functions. For $\lambda = 2$ they reduce to the elementary functions of §4.4.

Step 1: the change of variable. Let $t \equiv \tanh(\alpha x) \in (-1, 1)$ and $\psi(x) = w(t)$.

$$\frac{\mathrm{d}t}{\mathrm{d}x} = \frac{\alpha}{\cosh^2(\alpha x)} = \alpha(1 - t^2) \qquad \text{F1a,b}$$

$$\psi' = \frac{\mathrm{d}t}{\mathrm{d}x}\, w'(t) = \alpha(1 - t^2)\, w' \qquad \text{chain rule}$$

$$\psi'' = \frac{\mathrm{d}t}{\mathrm{d}x}\frac{\mathrm{d}}{\mathrm{d}t}\Big[\alpha(1 - t^2)w'\Big] = \alpha^2(1 - t^2)\Big[(1 - t^2)w'' - 2t\, w'\Big] \qquad \text{chain and product rules}$$

Insert these into (4.1), use $\frac{1}{\cosh^2(\alpha x)} = 1 - t^2$, and divide by $-\frac{\hbar^2\alpha^2}{2m}(1 - t^2)$:

$$(1 - t^2)\, w'' - 2t\, w' + \Big[\nu(\nu + 1) - \frac{\mu^2}{1 - t^2}\Big] w = 0, \qquad \nu(\nu + 1) = 2 \;\Rightarrow\; \nu = 1, \qquad \mu^2 \equiv -\frac{2mE}{\hbar^2\alpha^2}. \quad (4.17)$$

This is the associated Legendre equation, with *degree* $\nu = 1$ and *order* μ. (The other root, $\nu = -2$, gives the same equation.) The order depends on the energy:

$$E = -\frac{\hbar^2\kappa^2}{2m} < 0 :\; \mu = \frac{\kappa}{\alpha}; \qquad E = 0 :\; \mu = 0; \qquad E = \frac{\hbar^2 k^2}{2m} > 0 :\; \mu = \frac{ik}{\alpha} \text{ (imaginary)}.$$

For general λ, $\nu(\nu + 1) = \lambda(\lambda - 1)$ gives $\nu = \lambda - 1$. The walls $x = \pm\frac{L}{2}$ sit at $t = \pm\tanh(\alpha L/2)$, and the open line corresponds to $t \to \pm 1$ (C19).

Source. The equation: [DLMF] 14.2.2. The sech^2 well solved by an associated Legendre function of $\tanh x$: [LD, p. 3, Eqs. (3.8)–(3.10)]. Their $u'' + (E - \alpha\operatorname{sech}^2 x)\, u = 0$ is (4.1) in units $\hbar = 2m = 1$ with our $\alpha = 1$, if their constant α (not ours) is -2; their degree $\frac{1}{2}(\sqrt{1 - 4\alpha} - 1) = 1$ and order $i\sqrt{E}$ are our ν and μ.

Own step. (standard) The derivation above.

Step 2: the Legendre function of degree 1 in closed form. For $-1 < t < 1$ the (Ferrers) associated Legendre function is defined as

$$\mathsf{P}^{\mu}_{\nu}(t) = \Big(\frac{1 + t}{1 - t}\Big)^{\mu/2} \frac{F\big(\nu + 1, -\nu; 1 - \mu; \frac{1-t}{2}\big)}{\Gamma(1 - \mu)}.$$

For $\nu = 1$ the second parameter is $-\nu = -1$, so the series stops after one term (F15):

$$F\Big(2, -1; 1 - \mu; \frac{1 - t}{2}\Big) = 1 - \frac{2}{1 - \mu}\cdot\frac{1 - t}{2} = \frac{t - \mu}{1 - \mu}.$$

Dividing by $\Gamma(1 - \mu)$ and using $(1 - \mu)\Gamma(1 - \mu) = \Gamma(2 - \mu)$ (F14):

$$\mathsf{P}^{\mu}_{1}(t) = \Big(\frac{1 + t}{1 - t}\Big)^{\mu/2} \frac{t - \mu}{\Gamma(2 - \mu)}, \qquad \mathsf{P}^{-\mu}_{1}(t) = \Big(\frac{1 - t}{1 + t}\Big)^{\mu/2} \frac{t + \mu}{\Gamma(2 + \mu)}. \quad (4.18)$$

Because $1/\Gamma$ is finite everywhere (F14), (4.18) holds for every μ, including $\mu = 1$, where the factor $\frac{1}{1-\mu}$ alone would blow up. For $\mu = 0$ it gives $\mathsf{P}_1(t) = t$. Substituting (4.18) directly into (4.17) confirms that it is a solution, and it agrees with a library implementation for real and imaginary μ to 30 digits (C19).

Source. Definition: [DLMF] 14.3.1 with 14.3.3. $\mathsf{P}_1(t) = t$: [DLMF] 14.5.6.

Own step. The closed form (4.18) via the terminating series.

Step 3: a basis of solutions. Equation (4.17) is unchanged under $t \to -t$ (both w'' and $t\,w'$ keep their form), so $\mathsf{P}_1^{-\mu}(-t)$ is also a solution. Two solutions f_1, f_2 are independent when their Wronskian $f_1 f_2' - f_1' f_2$ is nonzero: otherwise the 2×2 system $c_1 f_1 + c_2 f_2 = 0$, $c_1 f_1' + c_2 f_2' = 0$ has a nonzero solution (c_1, c_2). Here

$$\mathcal{W}\{\mathsf{P}_1^{-\mu}(t),\, \mathsf{P}_1^{-\mu}(-t)\} = \frac{2}{\Gamma(\mu-1)\,\Gamma(\mu+2)\,(1-t^2)}.$$

By F14, $1/\Gamma(\mu-1) = 0$ exactly when $\mu - 1 \in \{0, -1, -2, \dots\}$; with $\operatorname{Re}\mu \geq 0$ that means $\mu = 1$ or $\mu = 0$ (C19).

- $\mu \neq 0, 1$ (every energy except $E = 0$ and $E = -\hbar^2\alpha^2/2m$): the general solution is $c_1\mathsf{P}_1^{-\mu}(t) + c_2\mathsf{P}_1^{-\mu}(-t)$.
- $\mu = 0$ ($E = 0$) and $\mu = 1$ ($E = -\hbar^2\alpha^2/2m$): the two functions coincide up to sign, and a second solution "of the second kind" is needed. For $\mu = 0$ it is $\mathsf{Q}_1(t) = \frac{t}{2}\ln\frac{1+t}{1-t} - 1$; for $\mu = 1$ it is y_2 of (4.10).

That two independent solutions give all solutions is the standard theorem that the solutions of a linear second-order ODE form a two-dimensional space (no page-checked reference). In this document it also follows from §4.3.

Source. Wronskian: [DLMF] 14.2.3 with $\nu = 1$. Q_1: [DLMF] 14.5.8.

Step 4: back to x, the same functions as before. By the definitions in F1, $\frac{1-t}{1+t} = \frac{\cosh\alpha x - \sinh\alpha x}{\cosh\alpha x + \sinh\alpha x} = e^{-2\alpha x}$, so $(\frac{1-t}{1+t})^{\mu/2} = e^{-\mu\alpha x}$. Then, from (4.18):

- $E < 0$, $\mu = \kappa/\alpha$. Using $e^{\mp\kappa x} = \cosh\kappa x \mp \sinh\kappa x$ in (4.9),

$$\alpha\,\Gamma(2+\tfrac{\kappa}{\alpha})\,\mathsf{P}_1^{-\kappa/\alpha}(\tanh\alpha x) = (\kappa + \alpha\tanh\alpha x)\,e^{-\kappa x} = \chi_e + \chi_o,$$
$$\alpha\,\Gamma(2+\tfrac{\kappa}{\alpha})\,\mathsf{P}_1^{-\kappa/\alpha}(-\tanh\alpha x) = (\kappa - \alpha\tanh\alpha x)\,e^{\kappa x} = \chi_e - \chi_o.$$

- $E > 0$, $\mu = ik/\alpha$. Using $e^{-ikx} = \cos kx - i\sin kx$ (F13),

$$\alpha\,\Gamma(2+\tfrac{ik}{\alpha})\,\mathsf{P}_1^{-ik/\alpha}(\tanh\alpha x) = (\alpha\tanh\alpha x + ik)\,e^{-ikx} = \psi_o + i\,\psi_e.$$

  The equation has real coefficients, so the real and imaginary parts, ψ_o and ψ_e of (4.8), are solutions themselves (F13).

- $E = 0$, $\mu = 0$: $\mathsf{P}_1 = \tanh\alpha x$ and, since $\ln\frac{1+t}{1-t} = 2\alpha x$, $\mathsf{Q}_1 = \alpha x\tanh\alpha x - 1$. These are the two solutions of §4.4(b).
- $E = -\hbar^2\alpha^2/2m$, $\mu = 1$: $\mathsf{P}_1^{-1}(t) = \frac{1}{2}\sqrt{1-t^2} = \frac{1}{2\cosh\alpha x} = \frac{1}{2}y_1$, and the second solution is y_2 of (4.10).

So both routes give identical functions (C19), and the box analysis of §4.5 applies unchanged.

Own step. Step 4.

Step 5: where the boundary conditions enter. A Legendre function by itself carries no boundary condition: it is just a solution of (4.17). The boundary condition picks the combination and the allowed energies.

- *Box (our problem):* $w = 0$ at $t = \pm\tanh(\alpha L/2)$. Since the functions are those of §4.4, this gives exactly the conditions (4.11) and (4.12).
- *Open line, $\psi \to 0$ as $x \to \pm\infty$, i.e. as $t \to \pm 1$, for $E < 0$ ($\mu > 0$):* as $t \to 1$, $\mathsf{P}_1^{-\mu}(t) \propto (1-t)^{\mu/2} \to 0$. As $t \to -1$, the factor $(\frac{1-t}{1+t})^{\mu/2}$ diverges, and the other factor tends to $t + \mu \to \mu - 1$. So $\mathsf{P}_1^{-\mu}(t) \to \infty$ unless $\mu = 1$; for $\mu = 1$ it is $\frac{1}{2}\sqrt{1-t^2} \to 0$. The partner $\mathsf{P}_1^{-\mu}(-t)$ is the mirror image. In a combination $c_1\mathsf{P}_1^{-\mu}(t) + c_2\mathsf{P}_1^{-\mu}(-t)$ that vanishes at both ends, the behaviour near $t = 1$ forces $c_2 = 0$ and near $t = -1$ forces $c_1 = 0$, unless $\mu = 1$ (C19). Hence the open line has exactly one bound state: $\mu = 1$, i.e. $E = -\hbar^2\alpha^2/2m$, $\psi \propto 1/\cosh(\alpha x)$, as in [CKS, p. 78].

Own step. Step 5.

4.8 Which route is better? A recommendation

	Factorization (SUSY), §4.1–4.5	Associated Legendre, §4.7
Mathematics needed	derivatives, product rule, linear maps (F3–F5)	also complex orders, Γ, hypergeometric series, Ferrers functions (F13–F15)
All solutions found?	yes, by a linear-algebra argument (§4.3)	yes, via a tabulated Wronskian plus the two-dimensional-solution-space theorem
Exceptional energies	one: $E = -\hbar^2\alpha^2/2m$	two: $E = 0$ and $E = -\hbar^2\alpha^2/2m$ (second-kind functions needed)
Phase shift, "reflectionless", one state removed	immediate: $A^\dagger$ maps the free wave to the shifted wave (4.14)	only after reducing $\mathsf{P}_1^{-\mu}$ to elementary form
Which λ	integer λ (the chain of partners ends at the free particle); $\lambda = 2$ needs one step	any λ, integer or not, from one formula ($\nu = \lambda - 1$)

General λ, for reference. For general λ the Legendre route gives open-line bound states at $\mu = \nu - n$, i.e. $E_n = -\frac{\hbar^2\alpha^2}{2m}(\lambda - 1 - n)^2$ for $n = 0, 1, \ldots$ with $n < \lambda - 1$. This is stated here without proof. It is checked by finite differences for $\lambda = 2,\ 2.5,\ 3$ (`C19`), and for $\lambda = 3$ it matches "$V_1 = -6\,\mathrm{sech}^2 x$ … two bound states at $E_1 = -4$ and $E_2 = -1$" [CKS, p. 78]. No page-checked source was found for general λ.

Recommendation (Claude's own judgement).

1. *For this project ($\lambda = 2$, hard walls), use the factorization (SUSY) route as the main derivation.*
   - It needs only Calculus 1 and linear algebra.
   - It proves within itself that all solutions have been found.
   - Its first output, $A^\dagger$ applied to a free wave, is already the phase-shifted wave (4.14). That gives $\delta(k)$ and the level density (4.16), which are exactly what Z_q needs in §5.
2. *Keep the Legendre route as an independent cross-check.* It reproduces every function (C19), so nothing is lost by not using it as the main line.
3. *Switch to the Legendre route if the project moves to non-integer λ*, i.e. to a continuously tunable well depth at fixed width.
   - Then the chain of partners no longer ends at the free particle (compare [CKS, pp. 41–42], integer case), so there are no elementary solutions, and the reflectionless property used in §4.6 is lost.
   - The box conditions must then be evaluated with $\mathsf{P}_\nu^{\pm\mu}(\pm t)$ numerically (for example `mpmath.legenp`).
   - For integer $\lambda > 2$ both routes work: SUSY needs $\lambda - 1$ factorization steps, while Legendre gives everything in one terminating formula.

Own step. The comparison and the recommendation.

5 The quantum partition function

By P3 and §4,

$$Z_q = e^{-\beta E_b} + \sum_{j \geq 1} e^{-\beta E_j}. \tag{5.1}$$

From here on, $\alpha\tanh(\alpha L/2)$ and κ_b are replaced by α (A3). The exact version keeps them; it is Eq. (5.7) below.

5.1 From the sum to an integral

Let $y(k) \equiv [kL - 2\arctan\frac{k}{\alpha}]/\pi$, so that $y(k_j) = j$ by (4.11). Then y is odd (F2a), $y(0) = 0$, and $y'(k) = [L - \frac{2\alpha}{k^2+\alpha^2}]/\pi > 0$. So y has a smooth, odd inverse $k(y)$, with $k'(y) = 1/y'(k(y))$ (inverse-function rule). Define

$$F(y) \equiv \exp\Big(-\frac{\beta\hbar^2 k(y)^2}{2m}\Big). \tag{5.2}$$

Then $F(j) = e^{-\beta E_j}$, F is smooth and even (because $k(-y)^2 = k(y)^2$), $F(0) = 1$, and F decays like a Gaussian. By F10,

$$\sum_{j\geq 1} e^{-\beta E_j} = \int_0^\infty F(y)\,\mathrm{d}y - \frac{1}{2} + R. \tag{5.3}$$

Own step. Applying F10 to this sum.

How small is R? (A4)

- *Empty box.* Here $F(y) = e^{-\beta E_1 y^2}$ with $\beta E_1 = \frac{\pi\lambda_{\rm th}^2}{4L^2}$ (using $\frac{\beta\hbar^2}{m} = \frac{\lambda_{\rm th}^2}{2\pi}$). Then (1.2) gives

$$R = \frac{2L}{\lambda_{\rm th}}\sum_{\ell\geq 1} e^{-4\pi\ell^2(L/\lambda_{\rm th})^2}: \qquad 7.0\times10^{-6},\ \ 5.9\times10^{-22},\ \ 4.6\times10^{-49} \ \text{ at } \ L/\lambda_{\rm th} = 1,\,2,\,3 \ (\texttt{C9a}). \tag{5.4}$$

- *Well + box*, computed from the exact levels at 80 digits (C9b): R equals (5.4) plus a part that is at most $4(\alpha L)^2 e^{-2\alpha L}$ times the bound-state weight $e^{a/2}$, in all nine cases computed ($\alpha L = 20, 30, 60$; $L/\lambda_{\rm th} = 1, 2, 3$).
- In the anomaly windows of §8, $L/\lambda_{\rm th} \geq 10$, where the empty-box part is below 10^{-500}. That dropping R is harmless there is also checked directly: $C_v^{\rm q}$ from the closed form agrees with the exact level sum to $2\times10^{-13}k_B$ (C13).

Own step. The size of R for the well. This is a numerical observation: estimating it analytically needs complex analysis, and no reference was found.

5.2 Change of variables, and the two integrals

Substitute $y = y(k)$, $\mathrm{d}y = y'(k)\,\mathrm{d}k$ (k runs from 0 to ∞ as y does), and use F6 and F11 with $s = \frac{\beta\hbar^2}{2m}$ and $c = \alpha$:

$$\begin{aligned}\int_0^\infty F(y)\,\mathrm{d}y &= \frac{L}{\pi}\int_0^\infty e^{-\beta\hbar^2k^2/2m}\,\mathrm{d}k - \frac{2\alpha}{\pi}\int_0^\infty \frac{e^{-\beta\hbar^2k^2/2m}}{k^2+\alpha^2}\,\mathrm{d}k \\ &= \frac{L}{\pi}\cdot\frac{1}{2}\sqrt{\frac{2\pi m}{\beta\hbar^2}} - \frac{2\alpha}{\pi}\cdot\frac{\pi}{2\alpha}\,e^{\beta\hbar^2\alpha^2/2m}\,\mathrm{erfc}\sqrt{\frac{\beta\hbar^2\alpha^2}{2m}} = \frac{L}{\lambda_{\rm th}} - e^{a/2}\,\mathrm{erfc}\,\sqrt{a/2}.\end{aligned} \tag{5.5}$$

The last step uses $\frac{L}{2\pi}\sqrt{\frac{2\pi m}{\beta\hbar^2}} = L\sqrt{\frac{m}{2\pi\hbar^2\beta}} = \frac{L}{\lambda_{\rm th}}$ and $\frac{\beta\hbar^2\alpha^2}{2m} = \frac{a}{2}$.

Own step. Eq. (5.5).

5.3 The closed form

Insert (5.3) and (5.5) into (5.1), with $e^{-\beta E_b} = e^{a/2}$, and use $1 - \mathrm{erfc} = \mathrm{erf}$:

$$Z_q = \frac{L}{\lambda_{\mathrm{th}}} - \frac{1}{2} + e^{a/2}\,\mathrm{erf}\,\sqrt{a/2}, \qquad a = \frac{\beta\hbar^2\alpha^2}{m}, \quad \frac{L}{\lambda_{\mathrm{th}}} = \frac{\alpha L}{\sqrt{2\pi a}}. \tag{5.6}$$

Exact version (A3 not used): keep $\alpha\tanh(\alpha L/2)$ in (5.5) and κ_b in the bound state,

$$Z_q = \frac{L}{\lambda_{\mathrm{th}}} - \frac{1}{2} + e^{\beta\hbar^2\kappa_b^2/2m} - e^{\beta\hbar^2\alpha^2\tanh^2(\alpha L/2)/2m}\,\mathrm{erfc}\,\sqrt{\frac{\beta\hbar^2\alpha^2\tanh^2(\alpha L/2)}{2m}} + R. \tag{5.7}$$

Without R, (5.7) and (5.6) both equal the direct level sum to 2×10^{-16} (relative) at $\alpha L = 100$ (C11). Replacing κ_b by α changes the bound-state exponent by $\frac{\beta\hbar^2(\alpha^2-\kappa_b^2)}{2m} \approx 4a\,e^{-\alpha L}$, by (4.13).

Own step. Eqs. (5.1)–(5.7). No reference found for this closed form.

Intuition

$$Z_q = \underbrace{\frac{L}{\lambda_{\mathrm{th}}}}_{\text{free particle}} \underbrace{-\frac{1}{2}}_{\text{hard walls}} + \underbrace{e^{a/2}}_{\text{bound state}} \underbrace{-\,e^{a/2}\,\mathrm{erfc}\,\sqrt{a/2}}_{\text{the continuum state the well removed}}.$$

- L/λ_{th}: the number of thermal wavelengths in the box (P5).
- $-\frac{1}{2}$: in (5.3), the integral includes the point $j=0$ with weight $F(0)/2 = \frac{1}{2}$ (Step 1 of F10), but $j=0$ ($k=0$) is not a level (§4.5 (ii)); the $-\frac{1}{2}$ removes it. Equivalently, $\frac{L}{\lambda_{\mathrm{th}}} - \frac{1}{2} = \frac{L-\lambda_{\mathrm{th}}/2}{\lambda_{\mathrm{th}}}$: each wall makes a layer $\lambda_{\mathrm{th}}/4$ unavailable.
- $e^{a/2} = e^{-\beta E_b}$: the bound state.
- The last term equals $\int_0^\infty \frac{1}{\pi}\frac{2\alpha}{k^2+\alpha^2}e^{-\beta\hbar^2k^2/2m}\mathrm{d}k$ (F11): the Boltzmann weight of the state removed in (4.16), which is spread over $k \lesssim \alpha$ (Fig. 1c). At high T ($a\to0$) it counts fully, tending to 1, and cancels the bound state: the well no longer matters. At low T ($a\to\infty$) only $\approx\sqrt{2/\pi a}$ of it is thermally reachable (F7d).

Own step. This reading.

6 The quantum heat capacity

Use P4(d) with $a = \frac{\hbar^2\alpha^2}{m}\beta$. Write $Z_q = B(a) - \frac{1}{2} + G(a)$, where $B \equiv \frac{L}{\lambda_{\mathrm{th}}} = \frac{\alpha L}{\sqrt{2\pi a}}$ and $G \equiv e^{a/2}\,\mathrm{erf}\,\sqrt{a/2}$. The derivatives ($' = \mathrm{d}/\mathrm{d}a$) are:

$$B \propto a^{-1/2}: \qquad B' = -\frac{B}{2a}, \qquad B'' = \frac{3B}{4a^2}, \tag{6.1}$$

$$G' = \frac{1}{2}e^{a/2}\,\mathrm{erf}\,\sqrt{a/2} + e^{a/2}\,\frac{e^{-a/2}}{\sqrt{2\pi a}} = \frac{G}{2} + \frac{1}{\sqrt{2\pi a}} \qquad \text{product rule, F7b} \tag{6.2}$$

$$G'' = \frac{G'}{2} - \frac{1}{2\sqrt{2\pi}}\,a^{-3/2}. \tag{6.3}$$

(C12)

$$\frac{C_v^{\mathrm{q}}}{k_B} = a^2\left[\frac{B''+G''}{B-\frac{1}{2}+G} - \left(\frac{B'+G'}{B-\frac{1}{2}+G}\right)^2\right], \qquad a = \frac{\beta\hbar^2\alpha^2}{m}. \tag{6.4}$$

Only αL and a appear, as (3.3) requires. For numerics, multiply Z_q, Z_q' and Z_q'' by $e^{-a/2}$; the ratios do not change and nothing overflows. Against $k_B\beta^2\mathrm{Var}(E)$ over the exact levels, (6.4) agrees to $2\times10^{-13}k_B$ for $\alpha L = 60, 100, 1000$ and $k_BT \in [0.02, 5]$ (C13).

Low T $(a \to \infty)$. Let $Z_c \equiv \sum_{j\ge1} e^{-\beta E_j} = \frac{L}{\lambda_{\mathrm{th}}} - \frac{1}{2} - e^{a/2}\,\mathrm{erfc}\,\sqrt{a/2}$ be the continuum part, so $Z_q = e^{a/2}(1+u)$ with $u = Z_c e^{-a/2} \ll 1$. Then $\ln Z_q = \frac{a}{2} + \ln(1+u) \approx \frac{a}{2} + u$, and since $Z_c \propto a^{-1/2}$ to leading order,

$$C_v^{\mathrm{q}} \approx k_B\, a^2 u'' = k_B\Big(\frac{a}{2}\Big)^2 Z_c\, e^{-a/2}\Big[1 + \frac{2}{a} + \dots\Big] \;\longrightarrow\; 0 \quad \text{(exponentially)}. \tag{6.5}$$

This is the two-level form P4(f), with gap $\Delta = \hbar^2\alpha^2/2m$ (so $\beta\Delta = a/2$) and the whole continuum acting as one excited level of "degeneracy" Z_c (C13: ratio 1.0151 at $a = 160$, against $1 + 2/a = 1.0125$).

High T $(a \to 0)$: see §8.2.

Own step. Eqs. (6.1)–(6.5).

7 The classical partition function and heat capacity

7.1 Partition function

By P5, over the box, with $-\beta V(x) = \frac{a}{\cosh^2(\alpha x)}$ from (3.2):

$$Z_{\mathrm{cl}} = \frac{1}{\lambda_{\mathrm{th}}} \int_{-L/2}^{L/2} e^{a/\cosh^2(\alpha x)}\,\mathrm{d}x. \tag{7.1}$$

Split off the flat part, writing $e^{a/\cosh^2} = 1 + (e^{a/\cosh^2} - 1)$, and extend the second integral to all x:

$$Z_{\mathrm{cl}} = \frac{L + I}{\lambda_{\mathrm{th}}} = \frac{\alpha L + \alpha I}{\sqrt{2\pi a}}, \qquad I \equiv \int_{-\infty}^{\infty} \Big(e^{a/\cosh^2(\alpha x)} - 1\Big)\mathrm{d}x. \tag{7.2}$$

Error of the extension (A3). For $|x| \ge \frac{L}{2}$, $v \equiv \frac{a}{\cosh^2(\alpha x)} \le v_0 \equiv 4ae^{-\alpha L}$ (F1e), and $e^v - 1 \le v\,e^{v_0}$. So the two added tails total at most $2e^{v_0}\frac{a}{\alpha}(1 - \tanh(\alpha L/2)) < \frac{4a}{\alpha}e^{-\alpha L}e^{v_0}$, which is $3\times10^{-42}/\alpha$ at $\alpha L = 100$, $a = 20$ (C14).

7.2 The well integral

Substitute $t = \tanh(\alpha x)$. Then $\mathrm{d}t = \frac{\alpha}{\cosh^2(\alpha x)}\mathrm{d}x = \alpha(1-t^2)\,\mathrm{d}x$ (F1a,b), $\frac{1}{\cosh^2(\alpha x)} = 1 - t^2$, and $x \in (-\infty, \infty)$ maps to $t \in (-1, 1)$:

$$\alpha I = \int_{-1}^{1} \frac{e^{a(1-t^2)} - 1}{1 - t^2}\,\mathrm{d}t. \tag{7.3}$$

The integrand is continuous on $[-1, 1]$, since $(e^{av} - 1)/v \to a$ as $v \to 0$.

Series. $\frac{e^{av}-1}{v} = \sum_{\ell\ge1} \frac{a^\ell v^{\ell-1}}{\ell!}$ converges uniformly for $v = 1 - t^2 \in [0, 1]$ (F9). Integrating term by term with F12 $(n = \ell - 1)$:

$$\alpha I = \sum_{\ell\ge1} \frac{a^\ell}{\ell!}\int_{-1}^{1}(1-t^2)^{\ell-1}\mathrm{d}t = \sum_{\ell\ge1}\frac{a^\ell}{\ell!}\cdot\frac{2^\ell(\ell-1)!}{(2\ell-1)!!} = \sum_{\ell\ge1}\frac{(2a)^\ell}{\ell\,(2\ell-1)!!} = 2a + \frac{2a^2}{3} + \frac{8a^3}{45} + \dots \tag{7.4}$$

It converges for every a: the ratio of successive terms is $\frac{2a\,\ell}{(\ell+1)(2\ell+1)} \to 0$. It agrees with (7.3) to 2×10^{-15} (C14).

7.3 Derivatives in a

By F8,

$$\alpha J_1 \equiv \frac{\mathrm{d}(\alpha I)}{\mathrm{d}a} = \int_{-1}^{1} e^{a(1-t^2)}\mathrm{d}t = 2e^a \int_0^1 e^{-at^2}\mathrm{d}t = \frac{2e^a}{\sqrt{a}}\int_0^{\sqrt{a}} e^{-u^2}\mathrm{d}u = \sqrt{\pi}\,\frac{e^a \operatorname{erf}\sqrt{a}}{\sqrt{a}}, \tag{7.5}$$

$$\alpha J_2 \equiv \frac{\mathrm{d}(\alpha J_1)}{\mathrm{d}a} = \int_{-1}^{1}(1-t^2)\,e^{a(1-t^2)}\mathrm{d}t = \alpha J_1\Big(1-\frac{1}{2a}\Big) + \frac{1}{a}. \tag{7.6}$$

In (7.5) the integrand is even, $u=\sqrt{a}\,t$, and F7 is used. The last form of (7.6) comes from differentiating (7.5) with the product rule and F7b: $\frac{\mathrm{d}}{\mathrm{d}a}\big[\sqrt{\pi}\,e^a a^{-1/2}\operatorname{erf}\sqrt{a}\big] = \alpha J_1(1-\frac{1}{2a}) + \sqrt{\pi}\,e^a a^{-1/2}\frac{e^{-a}}{\sqrt{\pi a}}$ (C12, C14). Undoing $t=\tanh(\alpha x)$,

$$J_1 = \int \frac{e^{a/\cosh^2(\alpha x)}}{\cosh^2(\alpha x)}\,\mathrm{d}x, \qquad J_2 = \int \frac{e^{a/\cosh^2(\alpha x)}}{\cosh^4(\alpha x)}\,\mathrm{d}x. \tag{7.7}$$

7.4 Heat capacity

From (7.2), $\ln Z_{\mathrm{cl}} = \ln(\alpha L+\alpha I) - \frac{1}{2}\ln(2\pi a)$. In P4(d): $a^2\frac{\mathrm{d}^2}{\mathrm{d}a^2}\big[-\frac{1}{2}\ln(2\pi a)\big] = a^2\cdot\frac{1}{2a^2} = \frac{1}{2}$, and $\frac{\mathrm{d}}{\mathrm{d}a}\ln(\alpha L+\alpha I) = \frac{\alpha J_1}{\alpha L+\alpha I}$, $\frac{\mathrm{d}^2}{\mathrm{d}a^2}\ln(\alpha L+\alpha I) = \frac{\alpha J_2}{\alpha L+\alpha I} - \Big(\frac{\alpha J_1}{\alpha L+\alpha I}\Big)^2$. Hence

$$\frac{C_v^{\mathrm{cl}}}{k_B} = \frac{1}{2} + a^2\left[\frac{\alpha J_2}{\alpha L+\alpha I} - \left(\frac{\alpha J_1}{\alpha L+\alpha I}\right)^2\right], \qquad \alpha I \text{ from (7.4)},\ \ \alpha J_1,\ \alpha J_2 \text{ from (7.5)–(7.6)}. \tag{7.8}$$

Intuition

This is (2.6). The $\frac{1}{2}$ is the kinetic energy (equipartition). By (7.7), $\langle 1/\cosh^2\rangle = J_1/(L+I)$ and $\langle 1/\cosh^4\rangle = J_2/(L+I)$ under $P(x)\propto e^{-\beta V(x)}$, so the bracket is $\mathrm{Var}(1/\cosh^2(\alpha x))$, and $a^2\,\mathrm{Var}(1/\cosh^2) = \beta^2\mathrm{Var}(V)$. Formula (7.8) agrees with a direct numerical integration of (2.6) to $10^{-9}k_B$ (C14).

Limits. *Low T* ($a\to\infty$): near $x=0$, $\frac{1}{\cosh^2(\alpha x)} = 1-\alpha^2x^2+O(x^4)$, so $V\approx -\frac{\hbar^2\alpha^2}{m} + \frac{\hbar^2\alpha^4}{m}x^2$, a harmonic oscillator with $\frac{1}{2}m\omega_0^2 = \frac{\hbar^2\alpha^4}{m}$. By equipartition (P5), $C_v^{\mathrm{cl}}\to k_B$; the next order is $C_v^{\mathrm{cl}}\approx k_B(1+1/a)$ (C14: 1.001004 at $a=1000$). In the same limit, $e^{a(1-t^2)}\approx e^a e^{-at^2}$ gives $\alpha I\approx e^a\sqrt{\pi/a}$ (F6). *High T* ($a\to 0$): $a^2\,\mathrm{Var}\to 0$, so $C_v^{\mathrm{cl}}\to k_B/2$.

Own step. Eqs. (7.1)–(7.8) and the limits, by standard techniques. No reference found for (7.4)–(7.8) for this potential.

8 Comparing C_v^{q} and C_v^{cl}

8.1 An exact relation between Z_q and Z_{cl}

By F7c with $z^2=a/2$ (so $2^n z^{2n}=a^n$), $G = e^{a/2}\operatorname{erf}\sqrt{a/2} = \sqrt{\frac{2a}{\pi}}\sum_{n\ge 0}\frac{a^n}{(2n+1)!!}$. By (7.4) with $\ell=n+1$, $\frac{I}{\lambda_{\mathrm{th}}} = \frac{\alpha I}{\sqrt{2\pi a}} = \sqrt{\frac{2a}{\pi}}\sum_{n\ge 0}\frac{2^n}{n+1}\frac{a^n}{(2n+1)!!}$. Subtracting, $Z_q - Z_{\mathrm{cl}} = -\frac{1}{2} + G - \frac{I}{\lambda_{\mathrm{th}}}$, so

$$Z_q = Z_{\mathrm{cl}} - \frac{1}{2} - D(a), \qquad D(a)\equiv \sqrt{\frac{2a}{\pi}}\sum_{n\ge 2}\Big(\frac{2^n}{n+1}-1\Big)\frac{a^n}{(2n+1)!!} = \sqrt{\frac{2a}{\pi}}\Big(\frac{a^2}{45}+\frac{a^3}{105}+\dots\Big) > 0. \tag{8.1}$$

The terms $n = 0, 1$ cancel exactly, and every coefficient with $n \geq 2$ is positive because $2^n > n + 1$ (C15). So at every T the quantum particle has *less* weight in the well than the classical one ($D > 0$, the zero-point cost), and the walls remove another $\frac{1}{2}$.

Own step. Eq. (8.1). No reference found.

8.2 High temperature

For $a \to 0$, $Z_{\rm cl} \approx L/\lambda_{\rm th} \gg \frac{1}{2} + D$, so $\ln Z_q - \ln Z_{\rm cl} = \ln\left(1 - \frac{1/2+D}{Z_{\rm cl}}\right) \approx -\frac{\lambda_{\rm th}}{2L} - \frac{\lambda_{\rm th}}{L} D$. Apply the sign rule P4(e) to each term:

- *Walls:* $-\frac{\lambda_{\rm th}}{2L} \propto \beta^{1/2}$, so $p = \frac{1}{2}$:

$$C_v^{\rm q} - C_v^{\rm cl} \xrightarrow{T\to\infty} k_B \frac{\lambda_{\rm th}}{8L} > 0. \tag{8.2}$$

- *Well:* $-\frac{\lambda_{\rm th}}{L} D \approx -\frac{\sqrt{2\pi a}}{\alpha L}\sqrt{\frac{2a}{\pi}}\frac{a^2}{45} = -\frac{2a^3}{45\,\alpha L} \propto \beta^3$, so $p = 3$, contributing $-\frac{4}{15}\frac{a^3}{\alpha L} k_B < 0$.

At high T the walls give the only excess, and the smooth well gives a deficit. At $k_B T = 2000\,\hbar^2\alpha^2/2m$ ($a = 0.001$) and $\alpha L = 100$: $C_v^{\rm q} - C_v^{\rm cl} = 9.9089 \times 10^{-5} k_B$, against $\lambda_{\rm th}/8L = 9.9083 \times 10^{-5}$ (C15).

Own step. This subsection.

8.3 The empty box: the wall effect alone

Without the well, (2.1) and (5.4) give $Z_q = y - \frac{1}{2} + R$ with $y \equiv L/\lambda_{\rm th} \propto \beta^{-1/2}$. Classically $Z_{\rm cl} = y$ exactly, so $C_v^{\rm cl} = k_B/2$ at every T. Drop R and use $\beta y' = -\frac{y}{2}$ and $\beta^2 y'' = \frac{3}{4} y$ ($' = \mathrm{d}/\mathrm{d}\beta$) in (2.3):

$$\frac{C_v^{\rm q}}{k_B} = \beta^2\left[\frac{y''}{y-\frac{1}{2}} - \left(\frac{y'}{y-\frac{1}{2}}\right)^2\right] = \frac{\frac{3}{4}y(y-\frac{1}{2}) - \frac{1}{4}y^2}{(y-\frac{1}{2})^2} = \frac{y(2y-\frac{3}{2})}{4(y-\frac{1}{2})^2},$$

and subtracting $\frac{1}{2}$:

$$\frac{C_v^{\rm q} - C_v^{\rm cl}}{k_B} = \frac{y-1}{8(y-\frac{1}{2})^2} = \frac{r(1-r)}{2(2-r)^2}, \qquad r \equiv \frac{\lambda_{\rm th}}{L} = \frac{1}{y}. \tag{8.3}$$

- It is positive iff $r < 1$, i.e. $L > \lambda_{\rm th}$, i.e. $k_B T > \frac{4}{\pi} E_1$ (using $r^2 = \frac{4}{\pi}\frac{E_1}{k_B T}$).
- $\frac{\mathrm{d}}{\mathrm{d}r}\frac{r(1-r)}{(2-r)^2} = \frac{2-3r}{(2-r)^3}$ vanishes at $r = \frac{2}{3}$, so the maximum is $\frac{(2/3)(1/3)}{2(4/3)^2} = \frac{1}{16}$, at $k_B T = \frac{9}{\pi} E_1 \approx 2.865 E_1$.
- As $T \to \infty$ it tends to $\frac{r}{8} = \frac{\lambda_{\rm th}}{8L}$, the wall term (8.2).

Against the exact level sum: the exact maximum is $0.562500 k_B$ at $2.864 E_1$, and the excess starts at $1.265 E_1$ (the formula gives $\frac{4}{\pi} = 1.273$). Formula (8.3) is good to 10^{-9} for $r < \frac{2}{3}$ (C16). Inside the anomaly windows below, $r \leq 0.1$, so this term is at most $0.013 k_B$.

Source. That $C_v^{\rm q} > C_v^{\rm cl}$ for a particle in a 1D box: [LGK, p. 95, Fig. 4.3 (right); p. 96].

Own step. Eq. (8.3) and its analysis.

Intuition

$Z_q = (L - \lambda_{\rm th}/2)/\lambda_{\rm th}$: the walls exclude a layer $\lambda_{\rm th}/4$ each. Heating shrinks the layer ($\lambda_{\rm th} \propto T^{-1/2}$) and gives the particle room, and that costs extra energy, hence extra heat capacity.

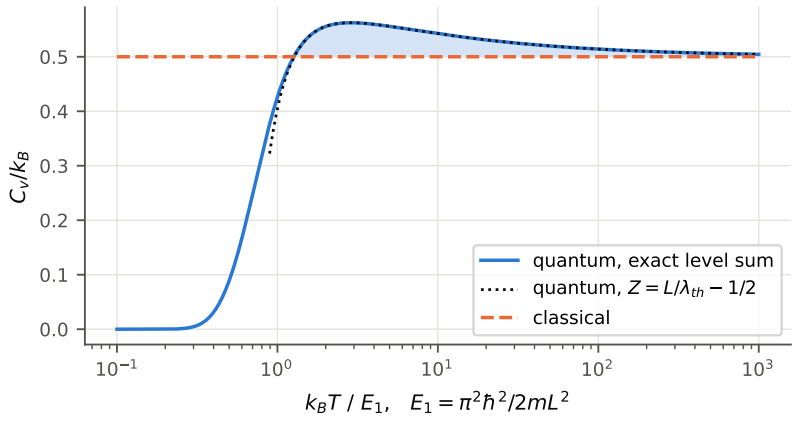


Figure 2: Empty box: exact level sum (blue), Eq. (8.3) (dotted) and the classical $k_B/2$ (orange, dashed), against k_BT/E_1.

8.4 The numbers

Table 2 evaluates the closed forms (6.4) and (7.8), with no further approximation, on 20 000 temperatures in $k_BT \in [0.02, 1]$ (C17). There is no closed-form proof here that $C_v^{\mathrm{q}} > C_v^{\mathrm{cl}}$ somewhere: the windows are found by evaluating the two formulas.

Table 2: Quantum against classical heat capacity. k_BT in units of $\hbar^2\alpha^2/2m$; "window" is where $C_v^{\mathrm{q}} > C_v^{\mathrm{cl}}$.

αL	$\max(C_v^{\mathrm{q}} - C_v^{\mathrm{cl}})/k_B$	at k_BT	C_v^{q}/k_B, C_v^{cl}/k_B there	window	C_v^{q} peak (at)	C_v^{cl} peak (at)	$\min(C_v^{\mathrm{q}} - C_v^{\mathrm{cl}})/k_B$ (at)
30	none for $k_BT < 1$	–	–	–	2.00 (0.393)	3.74 (0.412)	−1.75 (0.421)
70	−0.039 (none)	0.216	1.906, 1.946	–	3.04 (0.310)	5.40 (0.363)	−2.65 (0.386)
80	+0.095	0.214	2.093, 1.997	[0.195, 0.231]	3.23 (0.300)	5.69 (0.356)	−2.84 (0.379)
100	+0.343	0.210	2.427, 2.084	[0.176, 0.239]	3.56 (0.285)	6.21 (0.344)	−3.18 (0.367)
400	+2.591	0.184	5.091, 2.499	[0.129, 0.223]	6.05 (0.216)	10.03 (0.283)	−6.20 (0.299)
1000	+4.802	0.167	7.397, 2.595	[0.112, 0.203]	8.13 (0.185)	13.13 (0.252)	−9.10 (0.263)

8.5 Why there is a window (heuristic, not a proof)

Each C_v curve is dominated by one event: the particle leaving the well for the box (desorption). Treat it as a two-level system, P4(f), with "bound" and "free" weights:

- quantum: bound $e^{a/2}$ $(= e^{-\beta E_b})$, free L/λ_{th};
- classical: bound $I/\lambda_{\mathrm{th}} \approx \frac{e^a}{\sqrt{2}\,a}$ (from $\alpha I \approx e^a\sqrt{\pi/a}$ and $\alpha\lambda_{\mathrm{th}} = \sqrt{2\pi a}$), free L/λ_{th}. Here $e^a = e^{-\beta V_{\min}}$ and $\frac{1}{\sqrt{2}\,a} = \frac{k_BT}{\hbar\omega_0}$ is the oscillator partition function of the bottom [Tong-SP, p. 47, Eq. (2.20)], with $\hbar\omega_0 = \sqrt{2}\,\hbar^2\alpha^2/m$.

Desorption happens roughly where the weights are equal:

$$\text{quantum: } \frac{a_q}{2} = \ln\frac{\alpha L}{\sqrt{2\pi a_q}}, \qquad \text{classical: } a_{\mathrm{cl}} - \frac{1}{2}\ln a_{\mathrm{cl}} = \ln\frac{\alpha L}{\sqrt{\pi}}. \tag{8.4}$$

The ratio of bound weights, classical over quantum, is $\frac{e^{a/2}}{\sqrt{2}\,a} \approx 10$ at $a = 10$. The classical particle is held more strongly: it sits at the bottom $-\frac{\hbar^2\alpha^2}{m}$ and pays only the localization factor $k_BT/\hbar\omega_0$, whereas the zero-point energy puts the quantum bound state at $-\hbar^2\alpha^2/2m$, half the depth. Hence $T_q < T_{\mathrm{cl}}$. The

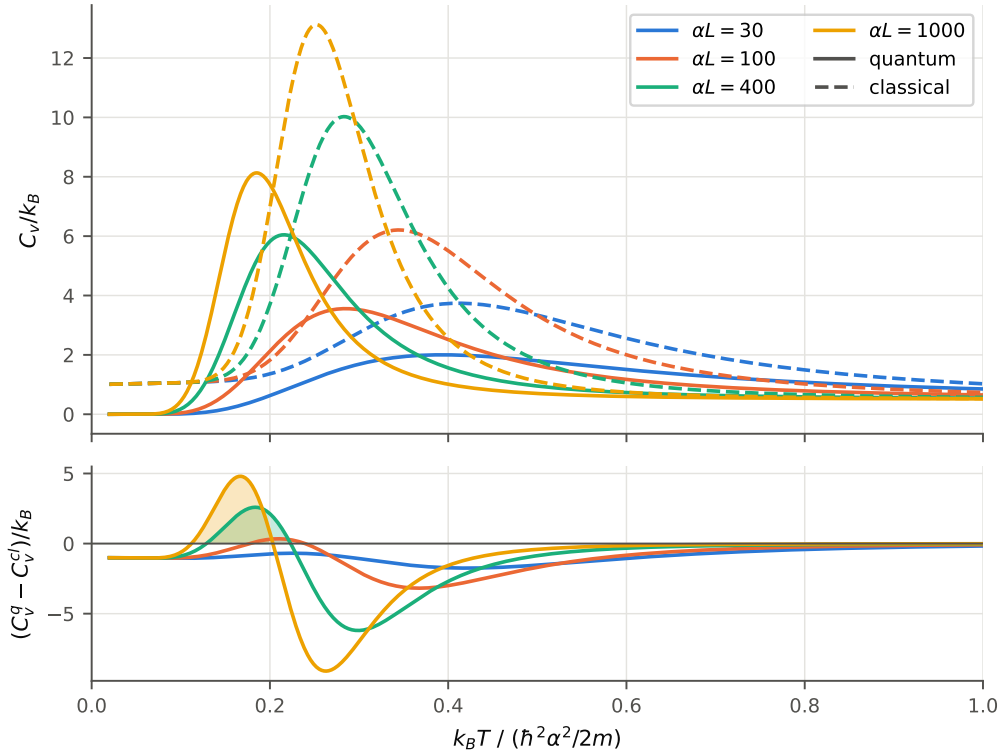


Figure 3: C_v^{q} (solid) and C_v^{cl} (dashed), and their difference, for four box lengths. Shaded: $C_v^{\mathrm{q}} > C_v^{\mathrm{cl}}$.

estimates (8.4) give $k_BT_q = 0.354, 0.248, 0.206$ and $k_BT_{\mathrm{cl}} = 0.415, 0.315, 0.273$ for $\alpha L = 100, 400, 1000$. The computed peaks are at 0.285, 0.216, 0.185 and 0.344, 0.283, 0.252: the estimates run 10–25% high, with the right order (C18).

Peak heights, from P4(f), $\approx (\beta\Delta)^2/4$:

- quantum: $\Delta_q = \hbar^2\alpha^2/2m + \frac{1}{2}k_BT$ (from the bound state to a gas with kinetic energy $\frac{1}{2}k_BT$);
- classical: $\Delta_{\mathrm{cl}} = \frac{\hbar^2\alpha^2}{m} - \frac{1}{2}k_BT$ (from the bottom plus oscillator energy k_BT to the gas).

This gives 4.0, 6.6, 8.7 (quantum) and 7.0, 10.8, 13.8 (classical), against the computed 3.56, 6.05, 8.13 and 6.21, 10.03, 13.13. The estimates run 5–15% high (C18).

Intuition

Below both peaks, $C_v^{\mathrm{q}} \approx 0$ (frozen in one state, (6.5)) while $C_v^{\mathrm{cl}} \approx k_B$ (vibrating at the bottom), so $C_v^{\mathrm{q}} < C_v^{\mathrm{cl}}$. The quantum particle desorbs first, so C_v^{q} climbs while C_v^{cl} is still about k_B: this is the window, $C_v^{\mathrm{q}} > C_v^{\mathrm{cl}}$. Then the classical desorption takes over, with an energy jump Δ about twice as large, and $C_v^{\mathrm{q}} < C_v^{\mathrm{cl}}$ again. A longer box (larger αL) gives more room (L/λ_{th}), so desorption happens at lower T, $\beta\Delta$ is larger, the peaks are sharper, and the window's excess grows. The window opens between $\alpha L = 70$ and 80.

Own step. The whole of §8.5: a heuristic, not a proof. No reference found.

8.6 What is exact and what is approximated

- *Exact:* the levels (4.11)–(4.12); Z_q in the form (5.7), including R; Z_{cl} in the form (7.1).
- *Dropped:* terms of relative size $e^{-\alpha L}$ (A3), and R (A4).

- *Checked* against exact level sums or direct integration: (C11), (C13), (C14).

A List of own steps

All steps below were constructed for this project by Claude. "(standard)" marks write-ups of textbook arguments.

Where	Step	Reference status
F1e–f, F3 lemma, F4, F5, F6 proof, F10 steps 1–2, F12 recursion	proofs of standard facts	(standard); results cited where a page-checked source exists
F11 proof	ODE in s for the Gaussian–Lorentzian integral	result: [DLMF] 7.7.1; route: none found
P1 example, (2.2), P4(a)–(f), (2.6)	standard consequences	(standard)
§3	the dimensional argument, (3.4)	(standard)
§4.3	all solutions are $A^{\dagger}$(free solutions)	extends [CKS, p. 14]
§4.4	explicit solutions; the case $E = -\hbar^2\alpha^2/2m$	none found
§4.5	box conditions, root counting, bound state, (4.13)	none found
§4.6	phase-shift reading, level slide, removed-state density	concepts: [Tong-Topics, p. 199], [CKS]
F13–F14	real/imaginary parts; $\Gamma(z+1) = z\Gamma(z)$ by parts	(standard)
§4.7	derivation of the Legendre form, closed form (4.18), steps 4–5	equation: [DLMF] 14.2.2; form: [LD, p. 3]
§4.8	comparison and recommendation	Claude's judgement
§5	sum to integral, R, change of variables, (5.6), (5.7), reading	none found
§6	(6.1)–(6.5)	none found
§7	(7.1)–(7.8), limits	none found
§8	(8.1), high-T terms, (8.3), heuristics	box effect shown in [LGK, p. 95]

B What `check_explained.py` verifies

C1	hyperbolic identities (F1)
C2, C3	factorization (4.5) and intertwining (4.6), symbolically for a generic f
C4, C5	(4.8), (4.9), (4.10) solve (4.1); they are $A^{\dagger}$ of the free bases; $Ay_1 = 0$
C6, C7	levels (4.11), Table 1, the threshold $\alpha L = 2.40$; bound state (4.12)–(4.13) at 40 digits
C8	far field (4.14); area of the removed-state density (4.16)
C9	R: empty box (5.4); well + box at 80 digits; the Bernoulli bound of F10
C10	F6, F11, F12 by quadrature
C11	(5.6) and (5.7) against the direct level sum
C12	derivatives (6.1)–(6.3), (7.6)
C13	(6.4) against $k_B\beta^2\mathrm{Var}(E)$ over the exact levels; the low-T form (6.5)
C14	(7.3), (7.4), (7.5), (7.6), (7.8) against quadrature; classical limits
C15	series of G and I/λ_{th}, $D > 0$, the high-T terms
C16	empty box: (8.3) against the exact level sum
C17, C18	Table 2; the heuristics of §8.5; sign rule P4(e)
C19	§4.7: the t-substitution; (4.18) solves (4.17) and matches `mpmath.legenp`; the Wronskian; the map to $\chi_e \pm \chi_o$ and $\psi_o + i\psi_e$; open-line regularity; general-λ bound states by finite differences

References

[Tong-QM] D. Tong, *Lectures on Quantum Mechanics* (Cambridge Part IB lecture notes). https://www.damtp.cam.ac.uk/user/tong/qm/qm.pdf; pages checked on the PDF dated 3 April 2026. *Pages used:* 9 (Eq. (1.1)); 10 (Eq. (1.2), H as a differential operator); 24–25 (infinite well: $\psi = 0$ where $V = \infty$, $\psi = 0$ at the walls, $k = n\pi/L$, Eq. (2.10), n and $-n$ the same state); 49–50 (bound states non-degenerate, via the Wronskian); 68 ($H\psi = E\psi$ as the analogue of $Au = \lambda u$).

[Tong-Topics] D. Tong, *Topics in Quantum Mechanics* (Cambridge Part II lecture notes). https://davidtong.org/pdfs/teaching/topics-in-quantum-mechanics/topicsinqm.pdf; pages checked on the PDF dated 14 January 2026. *Pages used:* 196 (parity: for $V(x) = V(-x)$, eigenstates can be chosen even or odd); 199 (S-matrix in the parity basis, $S_{++} = t + r$, $S_{--} = t - r$, phase shifts $\delta_{\pm}(k)$).

[Tong-SP] D. Tong, *Statistical Physics* (Cambridge Part II lecture notes). https://www.damtp.cam.ac.uk/user/tong/statphys/sp.pdf; pages checked on the PDF dated 17 April 2021. *Pages used:* 13 (two-state system, Eq. (1.14), exponential fall-off from the gap); 18 (Eqs. (1.19)–(1.22)); 20 (Eqs. (1.25)–(1.27)); 32 (Eq. (2.1), the $1/h$ factor); 33–34 (§2.1.1, the one-dimensional classical Z from the quantum one); 35 (Gaussian integral, Eqs. (2.5)–(2.6)); 37 (§2.2.1, equipartition); 47 (Eq. (2.20), classical oscillator, $E_{\text{vib}} = k_B T$).

[CKS] F. Cooper, A. Khare, U. Sukhatme, "Supersymmetry and quantum mechanics," Phys. Rep. **251**, 267 (1995). Pages are those of the free arXiv:hep-th/9405029v2, https://arxiv.org/abs/hep-th/9405029v2. *Pages used:* 13 (Eqs. (3)–(4)); 14 (Eqs. (5)–(11)); 15 (Eqs. (12)–(14), the zero mode has no partner); 17 ("one fewer bound state"; Eqs. (26)–(30), transmission amplitudes); 18 (Eqs. (32)–(33)); 41–42 (Eqs. (150)–(153), reflectionless $n\,\mathrm{sech}^2$ family, $T_1(k)$); 77 (Eq. (312), "V_1 holds n bound states"); 78 (case $n = 1$: $V_1 = -2\,\mathrm{sech}^2 x$, $E_1 = -1$).

[DJdC] S. Deffner, C. Jarzynski, A. del Campo, "Classical and quantum shortcuts to adiabaticity for scale-invariant driving," Phys. Rev. X **4**, 021013 (2014), open access, https://doi.org/10.1103/PhysRevX.4.021013. Local copy in `docs/refs`. *Page used:* 021013-9 (Table I, "modified Pöschl–Teller" row).

[DLMF] *NIST Digital Library of Mathematical Functions*, https://dlmf.nist.gov; equation *X.Y.Z* is at `https://dlmf.nist.gov/X.Y.EZ`. *Equations used:* 1.5.22–24, 1.8.16, 1.9.68, 2.10.1, 4.14.3, 4.14.10, 4.23.9, 4.23.12, 4.23.18, 4.24.9, 4.28.1, 4.28.2, 4.28.4, 4.34.1–3, 4.35.11–12, 4.35.23–27, 5.2.1 (and the text after it), 5.2.4–5, 5.4.6, 5.5.1, 5.12.1, 7.2.1–2, 7.4.1, 7.6.1–2, 7.7.1, 7.12.1, 14.2.2–3, 14.3.1, 14.3.3, 14.5.6, 14.5.8, 15.2.1, 15.2.4, 24.2.11–12, 24.9.1, 24.9.8–9.

[LD] W.-D. Li, W.-S. Dai, "A duality in classical and quantum mechanics," arXiv:1909.01089v1 (2019), https://arxiv.org/abs/1909.01089v1. *Page used:* 3 (§3.2, Eqs. (3.8)–(3.10): $u'' + (E - \alpha\,\mathrm{sech}^2 x)u = 0$ is solved by an associated Legendre function of $\tanh x$, with degree $\frac{1}{2}(\sqrt{1 - 4\alpha} - 1)$ and order $i\sqrt{E}$).

[LGK] A. Levy, D. Gelbwaser-Klimovsky, "Quantum features and signatures of quantum thermal machines," in *Thermodynamics in the Quantum Regime*, Fund. Theor. Phys. **195**, 87 (Springer, 2018). Book pages as in the local copy `docs/refs/qvsclas book chapter (1).pdf`; free version arXiv:1803.05586v1. *Pages used:* 95 [arXiv pp. 6–7] (classical oscillator $C_v = k_B$ by equipartition; Fig. 4.3 [arXiv Fig. 3], box: quantum above classical); 96 [arXiv p. 7] (box discussion).